Self Financing of Fiscal Deficits under Active Monetary Policy

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Abstract

This thesis studies to what extent a debt-financed government stimulus check can help pay for itself. I do so in a non-Ricardian environment with price rigidities, where the stimulus check can contribute to its own financing by expanding the tax base and increasing inflation, thereby eroding outstanding public debt. Building on Angeletos, Lian, and Wolf 2024, I study how active monetary policy and anticipated stimulus checks affect the check's self-financing potential. The main theoretical contribution is to show that self-financing does not necessarily become more potent when fiscal adjustment is delayed, in contrast to the monotonicity result in Angeletos et al. 2024. Second, I show that self-financing can increase when the check is announced before it is implemented. Under a neutral monetary regime, self-financing increases monotonically with implementation delay. Under active monetary policy, however, this monotonicity can break, thus limiting the effect of delayed implementation on self-financing.

Keywords: fiscal deficits, self-financing, monetary policy
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1 Introduction

Say the government issues a deficit-financed stimulus check to households. How is this newly issued public debt repaid? The answer that may jump to mind is *fiscal adjustment*: at some point, the government must raise taxes or cut spending. In this thesis, however, I study how the stimulus check can help pay for itself through its effects on the economy. I refer to this as *self-financing*. More specifically, I study how active monetary policy and pre-announced checks interact with the economy and affect self-financing. This is relevant in environments where the central bank works to stabilize the economy and where households anticipate the transfer before it is implemented.

Self-financing can come from two sources: tax-base expansion (i.e. increased output) which increases tax revenue through a proportional income tax τ_y and unexpected inflation which erodes the real value of public debt. For the tax-base to expand, the stimulus check must raise household demand, and thus it requires a failure of Ricardian equivalence. Furthermore, the check-induced demand must be met by increased production (i.e. supply), so that taken together the check increases output.

I introduce non-Ricardian households following the perpetual-youth framework developed in Blanchard 1985. Households are finitely lived and survive from one period to the next with constant probability. They therefore discount future taxes not only because of time preference, but also because of mortality risk. This makes them discount future tax (re-)payments more than the mechanical growth rate of debt - the real interest rate. A debt-financed transfer can thus increase demand. Firms experiencing price-setting rigidities, will partly meet this new demand by increasing production, and partly by increasing the price level. In an environment like this the stimulus check can expand the tax-base (thus increasing the tax revenues) and push up inflation (thus eroding public debt).

The thesis builds on work by Angeletos, Lian, and Wolf 2024, who study an overlapping generations new Keynesian (OLG-NK) model with perpetual-youth households and a social fund that makes aggregation analytically tractable. Otherwise the households follow a very standard utility function. The supply side consists of monopolistically competitive intermediary good firms with staggered price setting (Calvo 1983) and an aggregating perfectly competitive final good firm. Finally the government consists of a monetary authority setting the nominal interest rate, and a fiscal authority issuing one period bonds and setting taxes.

My first contribution is to characterize the equilibrium under a Taylor-like real-rate rule (Taylor 1993). Using this I study and compare the effects of fiscal adjustment on self-financing under neutral and hawkish monetary regimes.

Under a neutral real rate, lowering fiscal adjustment monotonically raises self-financing (as in Angeletos et al. 2024). Intuitively as the fiscal adjustment is delayed further, the households - which may not be alive to repay the debt - perceive a larger transfer. Thus the demand-led boom becomes larger, and with it, the potency of both self-financing channels. This is the self-proclaimed main result of Angeletos et al. 2024.

However, under a hawkish monetary policy rule, I show that this monotonicity can break. Firstly, higher rates increase the servicing costs of public debt. This also increases the persistence of debt due to roll-over costs. This increases the period over which debt servicing costs must be paid. Third, the output boom becomes more spread out, relative to the neutral case, because of the intertemporal substitution effects of the increased rate. Taken together, these adverse effects can cause self-financing to decrease when lowering the fiscal adjustment (in some range).

I then study the mechanism quantitatively under an empirically disciplined calibration. As in Angeletos et al. 2024, I augment the model with completely credit-constrained (HtM) households partly to match MPC evidence and compare directly with their results. Under a neutral real rate, the model replicates large self-financing shares, mostly through the tax-base channel. Under active monetary policy, however, the output boom is dampened, debt-service costs rise, and the maximum self-financing share is substantially lower.

Finally, I extend the analysis from surprise transfers to announced transfers. This matters because expected future transfers can affect output, inflation, and debt already at the announcement date. Under a neutral real rate, implementation delay raises self-financing by generating tax revenues before the transfer is paid and by lowering the present value of the transfer cost. Under active monetary policy, however, the effect of delay is more ambiguous. If the transfer is implemented (sufficiently) far in the future, so households discount it heavily, the direct demand effect at announcement is weak. At the same time, forward-looking firms raise prices already at the announcement date, prompting higher real interest rates under active monetary policy. This can reduce current demand and will raise debt-servicing costs. Hence, delaying implementation no longer necessarily increases self-financing.

Taken together, the thesis shows how the self-financing of a stimulus check is sensitive to both the monetary regime and pre-announcement of the transfer.

2 Model

The model closely follows Angeletos et al. 2024. I extend the model and their main analytical results to a more general Taylor-like monetary-policy rule.

2.1 Households

The economy features a unit continuum of perpetual youth households, $i \in (0,1]$ (see Blanchard 1985 and Yaari 1965) with constant survival probability $\omega \in [0,1]$. Hence, the probability of surviving $k \geq 0$ periods ahead is ω^k. The share of households dying in each period $(1-\omega)$ is replaced by newborns (with the same index) holding no assets. Newborns are further indexed with $h = n$ and old (surviving) households with $h = o$.

Households have a time-separable utility function and are not altruistic toward future generations. Hence they derive no utility from leaving bequests. Formally the utility of a household i alive at date t is given by

$$\mathrm{U}_{i,t} \equiv \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k\left(\frac{C_{i,t+k}^{1-1/\sigma}-1}{1-\frac{1}{\sigma}} - \frac{\iota L_{i,t+k}^{1+1/\varphi}}{1+\frac{1}{\varphi}}\right)\right], \quad \sigma \neq 1.^{1} \tag{1}$$

Here $C_{i,t} > 0$ denotes consumption and $L_{i,t} \geq 0$ denotes labour supply. The parameter $\beta \in (0,1)$ is the future discount factor, $\sigma > 0$ governs the intertemporal elasticity of substitution (IES) for consumption, $\varphi > 0$ is the Frisch elasticity of labour supply and $\iota > 0$ is a scale parameter governing the dis-utility of labour.

Households receive income from supplying labour, $W_{i,t}L_{i,t}$, from firm dividends, $Q_{i,t}$ and by investing in actuarially fair annuities, $A_{i,t}$. They have a return of I_t/ω - that is the risk-free return on one-period government bonds, I_t, scaled by the expected surviving population. Furthermore old households transfer funds $S^o_{i,t} = S^o$ to a social fund, paying out to newborns $S^n_{i,t} = S^n$. Formally the household budget constraint for period t is given by

$$A_{i,t+1} = \underbrace{\frac{I_t}{\omega}}_{\text{Annuity}}\Big(A_{i,t} + P_t(\underbrace{W_{i,t}L_{i,t} + Q_{i,t}}_{\equiv Y_{i,t}} - C_{i,t} - T_{i,t} + S_{i,t})\Big), \tag{2}$$

where $A_{i,t}$ denotes i's savings at the beginning of period t in nominal terms. Taxes $T_{i,t}$ are set according to a tax policy function $T_{i,t} = \mathrm{T}(Y_{i,t}, Z_t)$, where $Y_{i,t} \equiv W_{i,t}L_{i,t} + Q_{i,t}$ and Z_t denotes aggregate measures (for example outstanding real government debt).

The social fund is required to balance and we thus have $(1-\omega)S^n + \omega S^o =$

[1]For $\sigma = 1$ replace the consumption term with $\ln(C_{i,t})$.

$0 \iff S^{old} = -D^*(1-\omega)/\omega$, where D^* is steady-state (SS) public debt. And along with market clearing $A^* \equiv \int_0^1 A^* \, di = D^*$ in SS we have $\omega A_o^* + (1-\omega)A_n^* = D^* \iff A_o^* = D^*/\omega \implies A_o^* + S^o = D^* = S^n$. So the social fund ensures that all households receive the same assets (inclusive of the social fund) in SS.

To focus the analysis we abstract from heterogeneous labour supply and assume a union that demands $L_{i,t} = L_t$ by setting the wages to maximise the average utility of workers:

$$\max_{L_{i,t}} \int_0^1 \mathrm{U}_{i,t} \, di. \;\; s.t \;\; (2). \tag{3}$$

Note that we need to specify the tax function T which will be discussed in section 3.2. For now it is enough to know that τ_y is a proportional income tax and that there are no other distortionary taxes. This yields the sufficient first order condition (4) which sets the disposable income to match the average marginal rate of substitution between labour and consumption (see A.5).

$$(1-\tau_y)W_t = \frac{\iota L_t^{1/\varphi}}{\int_0^1 C_{i,t}^{-1/\sigma} di}. \tag{4}$$

With labour supply already chosen, household maximise (1) by choosing consumption and savings subject to (2). We obtain the standard necessary Euler equation (see A.1).

$$C_{i,t}^{-1/\sigma} = \beta\omega \frac{I_t}{\omega} \mathbb{E}_t \left[\frac{1}{\Pi_{t+1}} C_{i,t+1}^{-1/\sigma} \right]. \tag{5}$$

Here, the role of the annuity becomes clear: the discounting of future consumption due to mortality risk is offset by a larger return on savings.

Together, our assumptions and derivations imply some further simplifications. Specifically $Y_{i,t} = W_t L_t + Q_t = Y_t$, where we also assume dividend payments are equal across households, and $T_{i,t} = \mathrm{T}(Y_t, Z_t) = T_t$. Along with the social fund ensuring $A^* + S_{i,t} = D^*$ this implies that both cohorts $h \in \{o, n\}$ coincide in steady state: same income, labour supply, taxes, social fund adjusted assets and thus consumption (see A.3 for more detail). This will be useful in section 3 where I linearize the model around a steady-state.

2.2 Firms

The production side follows the standard New Keynesian setup, as in Galí 2015. A continuum of monopolistically competitive intermediate-good firms $j \in [0,1]$ produce differentiated goods using the linear technology $Y_{j,t} = AN_{j,t}$, where A is exogenous TFP. These differentiated goods are aggregated by a perfectly competitive final-good firm.

2.2.1 Final-good firm

The final-good firm chooses intermediate inputs to maximize profits, taking all prices as given:

$$\max_{\{Y_{j,t}\}_{j\in[0,1]}} P_t Y_t - \int_0^1 P_{j,t} Y_{j,t}\, dj \quad \text{s.t.} \quad Y_t = \left(\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}}\, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}. \tag{6}$$

The first-order condition gives the demand for each intermediate variety (see A.3)

$$Y_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t, \quad j \in [0,1]. \tag{7}$$

The associated aggregate price index is obtained by utilizing the zero-profit condition of competitive final-good firms $P_t Y_t - \int_0^1 P_{j,t} Y_{j,t}\, dj = 0$. Solving this condition for P_t yields (see B.2):

$$P_t = \left(\int_0^1 P_{j,t}^{1-\varepsilon}\, dj \right)^{\frac{1}{1-\varepsilon}}. \tag{8}$$

Thus, the final-good firm determines the demand curve faced by each intermediate-good producer and the price index.

2.2.2 Intermediate-good firms

Intermediate-good firms set prices subject to staggered price setting (Calvo 1983): in each period, a firm can re-optimize its price with probability $1-\theta$, independently of how long its current price has been in place. Since firms all face the same situation they choose a common reset price P_t^* when able to re-optimize. The aggregate price index then evolves according to

$$P_t^{1-\varepsilon} = (1-\theta)(P_t^*)^{1-\varepsilon} + \theta P_{t-1}^{1-\varepsilon}. \tag{9}$$

A firm that re-optimizes in period t chooses P_t^* to maximize expected discounted profits over the periods in which that price is expected to remain in effect (see B.3):

$$\max_{P_t^*} \mathbb{E}_t \left[\sum_{k=0}^{\infty} \theta^k \Lambda_{t,t+k} \Big(P_t^* Y_{t+k|t} - \Psi_{t+k}(Y_{t+k|t}) \Big) \right] \quad \text{s.t.} \quad Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k}, \tag{10}$$

where $\Lambda_{t,t+k} = \beta^k (C_{t+k}/C_t)^{-1/\sigma} P_t / P_{t+k}$ is the stochastic discount factor (derived from the Euler equation 5), $\Psi(\cdot)$ is the nominal cost function, and $Y_{t+k|t}$ denotes output in period $t+k$ by a firm that last reset its price in period t.

The optimal reset price satisfies

$$\mathbb{E}_t\Big[\sum_{k=0}^{\infty}\theta^k Q_{t,t+k}Y_{t+k|t}\left(\frac{P_t^*}{P_{t-1}}-\mathrm{M}MC_{t+k|t}\Pi_{t-1,t+k}\right)\Big]=0, \tag{11}$$

where $\mathrm{M}\equiv\varepsilon/(\varepsilon-1)$ is the markup in the corresponding frictionless environment, $MC_{t+k|t}$ is real marginal cost in period $t+k$ for a firm that reset its price in period t, and $\Pi_{t-1,t+k}\equiv P_{t+k}/P_{t-1}$.

Hence, the firm block pins down the model's supply side. The final-good firm determines demand for each differentiated variety, while intermediate-good firms choose production and set prices subject to Calvo rigidities. Together with the price index, production technology and wage setting, these conditions yield the New Keynesian Phillips curve derived in section 3.5.

2.3 Government

The government has two blocks: A fiscal authority that sets taxes and issues one-period riskless nominal debt, and a monetary authority that sets the nominal interest rate.

2.3.1 Fiscal authority

Let B_t denote beginning-of-period nominal public debt held by households. The fiscal authority's flow budget constraint is

$$\frac{B_{t+1}}{I_t}=B_t-P_tT_t, \tag{12}$$

where $T_t\equiv\int_0^1 T_{i,t}\,di$ is aggregate tax revenue and $1/I_t$ is the price of a bond that pays out 1 at $t+1$ - thus I_t is the gross return. Defining $D_t\equiv B_t/P_t$ and $R_t\equiv I_t/\mathbb{E}_t[\Pi_{t+1}]$, this can be written in real terms as (see C.1)

$$D_{t+1}=R_t(D_t-T_t)\left(\frac{\mathbb{E}_t[\Pi_{t+1}]}{\Pi_{t+1}}\right). \tag{13}$$

This expression highlights the inflation-erosion channel: unexpected inflation between t and $t+1$ reduces the real value of outstanding nominal debt.

Fiscal policy follows the tax rule in Angeletos et al. 2024:

$$T_{i,t}=\tau_yY_{i,t}+\overline{T}-\mathrm{E}_t+\tau_d(D_t-D^*+\mathrm{E}_t), \tag{14}$$

where $\overline{T}=T^*-\tau_yY^*$ balances the steady-state budget, E_t is a mean-zero i.i.d. deficit shock, and $\tau_d\in(0,1)$ governs the speed of fiscal adjustment. The parameter $\tau_y \in [0, 1)$ captures the tax-base channel: when output rises after a transfer, tax revenue rises mechanically.

2.3.2 Monetary authority

The monetary authority sets the nominal interest rate according to

$$\frac{I_t}{\Pi^*} = R^* \left(\frac{\Pi_t}{\Pi^*}\right)^{\psi} \left(\frac{Y_t}{Y^*}\right)^{\phi}. \tag{15}$$

The parameters $\psi, \phi \in \mathbb{R}$ determine how policy responds to inflation and output. As I will show in section 3.6 a deficit shock raises output and thus ψ and ϕ also determines how accommodative or hawkish monetary policy is toward fiscal expansions, where larger parameters imply more hawkish regimes. Note that the focus of this thesis will be on hawkish monetary rules.[2]

3 Equilibrium

Now I characterize the model's equilibrium analytically. To do so, I linearise the model equations around a deterministic steady state. Throughout, lower-case variables denote logarithmic deviations from their respective SS unless otherwise stated.

3.1 Linearisation

I linearise around the standard zero-inflation SS, $\Pi^* = 1$, where real allocations equal their flexible-price values and real public debt is constant at $D^* \geq 0$. The following results are therefore only valid in a small neighbourhood of this SS.

From the Euler equation (5) we have, $I^* = R^* = 1/\beta > 1$. From the real debt transition equation (13) we have, $T^* = (1 - \beta)D^*$ and as discussed in 2.1, social-fund-inclusive wealth satisfies $\tilde{A}^* = D^*$. Finally, since we have abstracted from government spending, goods market clearing implies $Y^* = C^*$. These SS relations are used throughout the linearization.

[2]In proposition 2 I offer a precise definition of hawkish monetary rules.

3.2 Law of Motion for Public Debt

(Log) Linearizing the fiscal tax rule (14) around the SS defined above yields the following equation (see C.1)

$$t_t = \underbrace{\tau_y y_t}_{\text{tax base}} + \underbrace{\tau_d (d_t + e_t)}_{\text{fiscal adjustment}} - e_t, \tag{16}$$

where $t_t \equiv (T_t - T^*)/Y^*,\ e_t \equiv \mathrm{E}_t / Y^*$. Linearizing the government's budget constraint (13) and inserting the tax rule (16) we obtain the linearised law of motion for real public debt around SS (see C.1):

$$d_{t+1} = \frac{1}{\beta}\Big(d_t \underbrace{+ e_t - \tau_y y_t - \tau_d (d_t + e_t)}_{=-t_t}\Big) + \frac{D^*}{Y^*} r_t - \frac{D^*}{Y^*}(\pi_{t+1} - E_t[\pi_{t+1}]) \tag{17}$$

The first two terms are the expected debt for the next period, consisting of roll-over costs of debt and net tax payments. The last term captures debt erosion due to unexpected inflation.

Let initial nominal debt B_0 be predetermined such that (starting from SS in period -1) we have the initial condition for debt

$$d_0 = -\frac{D^*}{Y^*}\pi_0. \tag{18}$$

3.3 Monetary Policy

Linearizing the monetary policy rule (20) around SS yields (see C.2)

$$i_t = \psi \pi_t + \phi y_t, \tag{19}$$

Using the definition of the ex-ante real interest rate, $R_t \equiv I_t / \mathbb{E}_t[\Pi_{t+1}]$, its log-linear approximation is $r_t = i_t - \mathbb{E}_t[\pi_{t+1}]$ (see A.2) . Substituting (19) into this yields the real-rate monetary policy equation:

$$r_t = \psi \pi_t + \phi y_t - \mathbb{E}_t[\pi_{t+1}]. \tag{20}$$

This represents a substantial departure from Angeletos et al. 2024, where the real rate follows the rule $r_t = \phi y_t$. In Section 3.6, we show that their rule is nested in equilibrium.

3.4 Aggregate Demand

From section 2.1 we know that households share the same SS values. Hence, linearizing the consumption Euler equation (5) and the budget constraint around this common state gives (see A.2 and A.3)

$$\mathbb{E}_t[c^h_{i,t+1}] = c^h_{i,t} + \sigma r_t, \tag{21}$$

$$\tilde{a}^h_{i,t} = \beta\omega\mathbb{E}_t[\tilde{a}^h_{i,t+1}] + c^h_{i,t} - (y_{i,t} - t_t) - \beta\frac{D^*}{Y^*}r_t, \tag{22}$$

where $\tilde{a}^h_{i,t} \equiv (A^h_{i,t}/P_t + S_{i,t} - D^*)/Y^*$ is social-fund-inclusive real wealth in deviation from SS output.

Iterating (22) forward, imposing the no-Ponzi condition

$$\lim_{T\to\infty}(\beta\omega)^T\mathbb{E}_t[\tilde{a}^h_{i,t+T}] = 0,$$

and using (21) to substitute for future consumption gives the household consumption function (see A.4):

$$c^h_{i,t} = (1-\beta\omega)\left(\tilde{a}^h_{i,t} + \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k(y_{t+k} - t_{t+k})\right]\right) - \xi\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right], \tag{23}$$

where $\xi \equiv \sigma\beta\omega - (1-\beta\omega)\beta D^*/Y^*$, and $t_t \equiv (T_t - T^*)/Y^*$. Note that this simple aggregation is possible exactly because all households share the same SS, preferences and budget constraint. Here, the thesis also departs from Angeletos et al. 2024, who chose the monetary policy such that $r_t = 0$. The following derivations therefore nest their case while allowing for a non-neutral real-rate response.

Aggregating over households gives the economy's consumption function

$$c_t = (1-\beta\omega)\left(\tilde{a}_t + \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k(y_{t+k} - t_{t+k})\right]\right) - \xi\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right], \tag{24}$$

where $c_t \equiv \int_0^1 c^h_{i,t}\,di$ and $\tilde{a}_t \equiv \int_0^1 \tilde{a}^h_{i,t}\,di$. Because household assets equal public debt we also have, $\tilde{a}_t = d_t$.

Substituting the tax rule (16) and imposing market clearing, $c_t = y_t$, yields one of the main equations of the thesis (see D):

$$y_t = \mathrm{X}_d(d_t + e_t) + \mathrm{X}_y\mathbb{E}_t\left[(1-\beta\omega)\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right] - \mathrm{X}_r\mathbb{E}_t\left[(1-\beta\omega)\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right], \tag{25}$$

where $\mathrm{X}_d \equiv \frac{(1-\beta\omega)(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}$, $\mathrm{X}_y \equiv 1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}$, $\mathrm{X}_r \equiv \frac{\sigma\beta\omega}{1-\beta\omega} - \frac{D^*}{Y^*}\frac{\beta(1-\omega)}{1-\omega(1-\tau_d)}$.

Equation (25) is the model's aggregate demand curve. The first term captures the wealth effect of public debt. Since $\mathrm{X}_d > 0$ if and only if $\omega < 1$ and $\tau_d < 1$, debt affects demand only when Ricardian equivalence fails and fiscal adjustment is not immediate. Intuitively, current households value the transfer more than the future tax repayments because they discount future taxes at $(\beta\omega)^k$, while public debt rolls over at $R^* = \beta^{-1}$. A deficit shock therefore effectively transfers wealth from future generations (having to pay back the debt) to current ones.

In the infinitely lived benchmark, $\omega = 1$, this wealth effect disappears (i.e. $\mathrm{X}_d = 0$). In that case, (25) reduces to the textbook AD equation (see D.3),

$$y_t = -\sigma r_t + \mathbb{E}_t[y_{t+1}],$$

showing that debt affects demand only when households are finitely lived and therefore discount the future more heavily.

The second term is the *intertemporal Keynesian cross* (IKC). It captures the general-equilibrium feedback from income to demand. Thus, $\mathrm{X}_y \in (0, 1]$ measures the strength of the Keynesian multiplier.[3]

Finally, X_r measures the response to expected real rates. Its sign is ambiguous: higher expected rates reduce demand through intertemporal substitution, but can raise demand through a wealth effect when households hold financial assets, $D^* > 0$. With infinitely lived households, this wealth effect is exactly offset by future taxes. With $\omega < 1$, the offset is incomplete, so real rates affect demand through both substitution and wealth.

3.5 Aggregate Supply

The supply side follows from the price-setting problem of intermediate-good firms. Log-linearizing the reset-price condition (11) around the zero-inflation SS gives (see B.3 for detailed derivations of this section's equations)

$$p_t^* = (1-\theta\beta)\sum_{k=0}^{\infty}(\beta\theta)^k \mathbb{E}_t[mc_{t+k|t} + p_{t+k}]. \tag{26}$$

A firm that can reset its price therefore chooses a weighted average over its expected future marginal costs and the aggregate price index. The weights reflect discounting, β^k, and the probability that the price is still in place k periods ahead, θ^k.

Writing (26) recursively and subtracting p_t yields

$$p_t^* - p_t = (1-\theta\beta)mc_t + \beta\theta\mathbb{E}_t[p_{t+1}^* - p_{t+1}] + \beta\theta\mathbb{E}_t[\underbrace{p_{t+1} - p_t}_{\equiv\pi_{t+1}}], \tag{27}$$

[3] Let $\mathrm{D} \equiv 1-\omega(1-\tau_d) = 1-\omega+\omega\tau_d > 1-\omega$. Then $\tau_y(1-\omega)/\mathrm{D} < \tau_y < 1$, implying $\mathrm{X}_y \in (0, 1]$.

where linear technology implies that real marginal cost is common across firms, $mc_{t|t} = mc_t$.

Next, linearizing the aggregate price index (9) and rearranging gives $p_t^* - p_t = \pi_t\theta/(1-\theta)$. Combining this with (27) gives the New Keynesian Phillips curve (NKPC) in marginal-cost form:

$$\pi_t = \tilde{\kappa}\, mc_t + \beta \mathbb{E}_t[\pi_{t+1}] = \tilde{\kappa} \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t[mc_{t+k}], \tag{28}$$

where $\kappa \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}$. That is, inflation grows with expected future marginal costs. This is because, when marginal costs increase, or equivalently markups decrease below their desired level. Therefore resetting firms raise prices to restore markups pushing up inflation.

To express the NKPC in output, first log-linearise the labour supply condition (4) and impose market clearing, $c_t = y_t$:

$$w_t = m_w y_t,$$

where $m_w \equiv \frac{1}{\varphi} + \frac{1}{\sigma}$. With linear production, $y_t = n_t$, real marginal cost equals the real wage, so $mc_t = w_t$. Hence,

$$\pi_t = \kappa y_t + \beta \mathbb{E}_t[\pi_{t+1}], \tag{29}$$

where $\kappa \equiv \tilde{\kappa} m_w > 0$. The parameter κ is the slope of the NKPC: for given expected inflation, a larger κ means that a given output gap generates more inflation.

In the limiting case of fully flexible prices, $\theta \to 0^+$ (and $\kappa \to \infty$). Firms then reset prices immediately and preserve their desired markup, so demand expansions are absorbed by prices rather than real activity. Conversely, a flatter NKPC shifts the response toward output. Thus κ affects the composition of self-financing.

3.6 Equilibrium Definition and Properties

The equilibrium is defined in the standard way: households and firms optimize, markets clear, the government budget constraint holds, and the no-Ponzi restriction for the household and government budget constraints are respectively satisfied. These requirements are encapsulated in the following definition.

Definition 1 (Equilibrium path)**.** *An equilibrium path is a bounded stochastic path* $\{y_t, \pi_t, d_t\}_{t=0}^{\infty}$ *satisfying aggregate supply* (29)*, aggregate demand* (25)*, the monetary* *rule* (20)*, the debt law of motion* (17)*, the government no-Ponzi condition*

$$\lim_{k\to\infty} \mathbb{E}_t \left[\beta^k d_{t+k}\right] = 0,$$

and the initial debt condition (18)*.*[4]

Using the NKPC to substitute for expected inflation in the real-rate equation (20) gives

$$r_t = \alpha_y y_t + \alpha_\pi \pi_t, \tag{30}$$

where $\alpha_y \equiv \phi + \frac{\kappa}{\beta}$ and $\alpha_\pi \equiv \psi - \frac{1}{\beta}$. Note that $r_t = 0$ for $\phi = -\kappa/\beta$ and $\psi = 1/\beta$ and thus Angeletos et al. 2024 is nested in the parameter space. Combining (30) with the other equilibrium conditions yields the linear rational-expectations system

$$\mathbb{E}_t \begin{bmatrix} d_{t+1} \\ y_{t+1} \\ \pi_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} m_{q,q} & m_{q,y} & m_{q,\pi} \\ m_{y,d} & m_{y,y} & m_{y,\pi} \\ 0 & -\frac{\kappa}{\beta} & \frac{1}{\beta} \end{bmatrix}}_{\equiv A} \begin{bmatrix} d_t + e_t \\ y_t \\ \pi_t \end{bmatrix}, \tag{31}$$

where the coefficients $m_{i,j}$ are defined in E.2.

This thesis studies the unique bounded equilibrium path of the model, when it exists. Blanchard and Kahn 1980 give the requirements for such a path: the number of unstable eigenvalues (i.e. eigenvalues with modulus strictly greater than 1) of the transition matrix must equal the number of jump variables. Even though the system in (31) has no predetermined variables (remember that public debt, $d_t \equiv (B_t/P_t - D^*)/Y^*$ depends on the current price level P_t) one can show that the BK-condition is fulfilled when A has exactly two unstable eigenvalues:[5]

Lemma 1. *The system in* (31) *has a unique bounded equilibrium path if the eigenvalues of the transition matrix A fulfill $|\lambda_s| < 1 < |\lambda_{u1}|, |\lambda_{u2}|$.*

Proof. See appendix E.3 □

Applying lemma 1 we can characterize the unique equilibrium of system (31) (when it exists).

Proposition 1 (Unique bounded path)**.** *Assume that A in* (31) *has eigenvalues satisfying, $|\lambda_s| < 1 < |\lambda_{u1}|, |\lambda_{u2}|$. Then the unique bounded equilibrium path exists and follows*

$$\mathbb{E}_t[d_{t+1}] = \lambda_s (d_t + e_t), \qquad y_t = \chi(d_t + e_t), \qquad \pi_t = \eta(d_t + e_t), \tag{32}$$

[4] Boundedness follows Blanchard and Kahn 1980: system variables must not explode to $\pm\infty$ over time.

[5] This mirrors Angeletos et al. 2024 that require one unstable root of the 2×2 transition matrix of their system in $(d_t + e, y_t)$.

with the initial value of real public debt

$$d_0 = -\frac{\eta D^*/Y^*}{1+\eta D^*/Y^*} e_0, \tag{33}$$

and where the coefficients solving the fixed point relation

$$\begin{aligned} \chi &= X_d + \frac{1-\beta\omega}{1-\beta\omega\lambda_s}\left[X_y\chi - X_r\left(\left(\phi+\frac{\kappa}{\beta}\right)\chi + \left(\psi-\frac{1}{\beta}\right)\eta\right)\right], \\ \lambda_s &= \frac{1-\tau_d}{\beta} - \frac{\tau_y}{\beta}\chi + \frac{D^*}{Y^*}\left[\left(\phi+\frac{\kappa}{\beta}\right)\chi + \left(\psi-\frac{1}{\beta}\right)\eta\right], \quad \eta = \frac{\kappa\chi}{1-\beta\lambda_s}. \end{aligned} \tag{34}$$

Condition (32) describes the equilibrium path. It gives λ_s as the expected persistence of public debt and shows that χ and η measure the output and inflation response to this fiscal state respectively. Therefore λ_s also measures the persistence of the output and inflation responses.

The relation (34) contains the fixed point between aggregate demand, debt dynamics and inflation. The first relation shows how aggregate demand depends on the persistence λ_s, inflation η (through the monetary policy) and the strength of the IKC feedback mechanism. Note that inflation does not feed back into demand under the neutral rule.

The sign of χ is theoretically ambiuous, however, the next proposition gives a simple - and empirically relevant - sufficient condition under which $\chi > 0$.

Proposition 2 (Deficits trigger a boom under hawkish monetary policy)**.** *Assume $\omega < 1$, $\tau_d < 1$, $X_r \geq 0$ and monetary policy is hawkish in the sense that*

$$\alpha_y \equiv \phi + \frac{\kappa}{\beta} \geq 0, \qquad \alpha_\pi \equiv \psi - \frac{1}{\beta} \geq 0.$$

Then

$$\chi > 0.$$

Proof. See Appendix E.5. □

Hence, in the neutral-to-hawkish policy region considered in this thesis, deficits trigger output booms. Throughout this equilibrium analysis I assume the necessary condition for proposition 2 to hold. In short I assume that a deficit shock induces an output boom.

The second relation gives debt persistence. We see that faster fiscal adjustment and, for $\tau_y > 0$, larger output booms work to lower fiscal persistence. The last term is the monetary/debt-service channel, wich under a hawkish monetary regime increases debt-rollover costs and thus persistence.

The third relation pins down inflation as a function of the output boom and the persistence of the fiscal state. A larger output response raises marginal costs and therefore inflation. Persistence also matters: if λ_s is higher, firms expect marginal costs to remain elevated for longer, so inflation rises more already today. Since $1 - \beta\lambda_s > 0$, η has the same sign as χ. Thus, deficits also increase inflation.

Proposition 1 generalizes the fixed-point logic in Angeletos et al. 2024: Deficits increase demand; demand feeds back into debt through the tax base - which again affects demand. Inflation is then recovered afterwards.[6] The difference is that active monetary policy introduces an additional feedback: output and inflation can raise real rates, which affects household demand directly and public debt dynamics through debt-service costs. The equilibrium is therefore pinned down jointly by the demand response χ, debt persistence λ_s, and the inflation response η.

4 Self-financing of fiscal deficits

To study this formally, we need a quantitative measure. This section defines this measure and applies it to show how hawkish monetary policy can qualitatively change the equilibrium dynamics - compared to Angeletos et al. 2024.

4.1 The self-financing share

We decompose a deficit shock $e_0 > 0$ into sources of financing. Formally we iterate forward the debt transition equation (17) and apply period 0 expectations yielding

$$e_0 + \underbrace{\frac{D^*}{Y^*}\Big(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_0[r_k]\Big)}_{\text{debt servicing costs}} = \underbrace{\tau_d\Big(e_0 + \sum_{k=0}^{\infty}\beta^{k}\mathbb{E}_0[d_k]\Big)}_{\text{fiscal adjustment}} + \underbrace{\tau_y\Big(\sum_{k=0}^{\infty}\beta^{k}\mathbb{E}_0[y_k]\Big)}_{\text{taxbase expansion}} + \underbrace{\frac{D^*}{Y^*}(\pi_0 - \mathbb{E}_{-1}[\pi_0])}_{\text{period 0 inflation}}. \tag{35}$$

We see that the deficit shock can affect public debt through four channels. (i) Increased debt servicing costs due to the reaction of the hawkish monetary authority, (ii) fiscal adjustment due to increased taxes in response to larger debt, (iii) self-financing from tax-base expansion raising taxes proportional to τ_y and (iv) self-financing from period 0 inflation eroding the real debt. From this we define the self-financing share ν.

[6] This is why they analyse the simpler system in $d_t + e_t$ and y_t.

Definition 2. *The self-financing share ν is defined as* [7]

$$\nu \equiv \frac{\tau_y\left(\sum_{k=0}^{\infty}\beta^k\mathbb{E}_0[y_k]\right)+\frac{D^*}{Y^*}(\pi_0-\mathbb{E}_{-1}[\pi_0])}{e_0+\frac{D^*}{Y^*}\left(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_0[r_k]\right)} \tag{36}$$

Self-financing is thus the present value of the additional revenues generated by the tax-base expansion, together with the period-0 inflation erosion of public debt, relative to the direct cost of the transfer and any increase in debt-service costs following the deficit shock.

Note that this can be decomposed into self-financing from tax base expansion ν_y and from inflation ν_π. That is $\nu = \nu_y + \nu_\pi$, where

$$\nu_y \equiv \frac{\tau_y\left(\sum_{k=0}^{\infty}\beta^k\mathbb{E}_0[y_k]\right)}{e_0+\frac{D^*}{Y^*}\left(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_0[r_k]\right)} \quad \text{and} \quad \nu_\pi \equiv \frac{\frac{D^*}{Y^*}\pi_0}{e_0+\frac{D^*}{Y^*}\left(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_0[r_k]\right)}.$$

4.2 Fiscal adjustment and degree of self-financing

With definition 2 in place we can state the first theorem of the thesis regarding how ν changes with τ_d.

Theorem 1. *Assuming that χ defined in proposition 1 is differentiable in $\tau_d \in (0,1)$ then given a one-off deficit shock (i.e. $e_0 > 0$) the degree of self-financing ν satisfies*

$$\frac{\partial \nu}{\partial \tau_d} < 0 \iff \tau_d \frac{\partial \chi}{\partial \tau_d} < \chi, \tag{37}$$

which does not generally hold.

Proof. See appendix F.2 □

Hence, active monetary policy changes the self-financing result in a qualitative way. Theorem 1 shows the result in Angeletos et al. 2024, where lower τ_d monotonically raises self-financing, is not a general result. It relies on their neutral monetary rule.

To build intuition on the mechanisms behind this result, I compare the model outcomes under a neutral regime, $r_t = 0$, and a hawkish regime, respectively. Figure 1 shows the impulse response functions of a one-off deficit shock, $e_0 = \mathrm{E}_0/Y^* = 0.01$ (which we characterised in proposition 1) under neutral monetary policy.

[7] Note, for $\phi = -\kappa/\beta$ and $\psi = 1/\beta$ definition (2) reduces to the definition used in Angeletos et al. 2024.

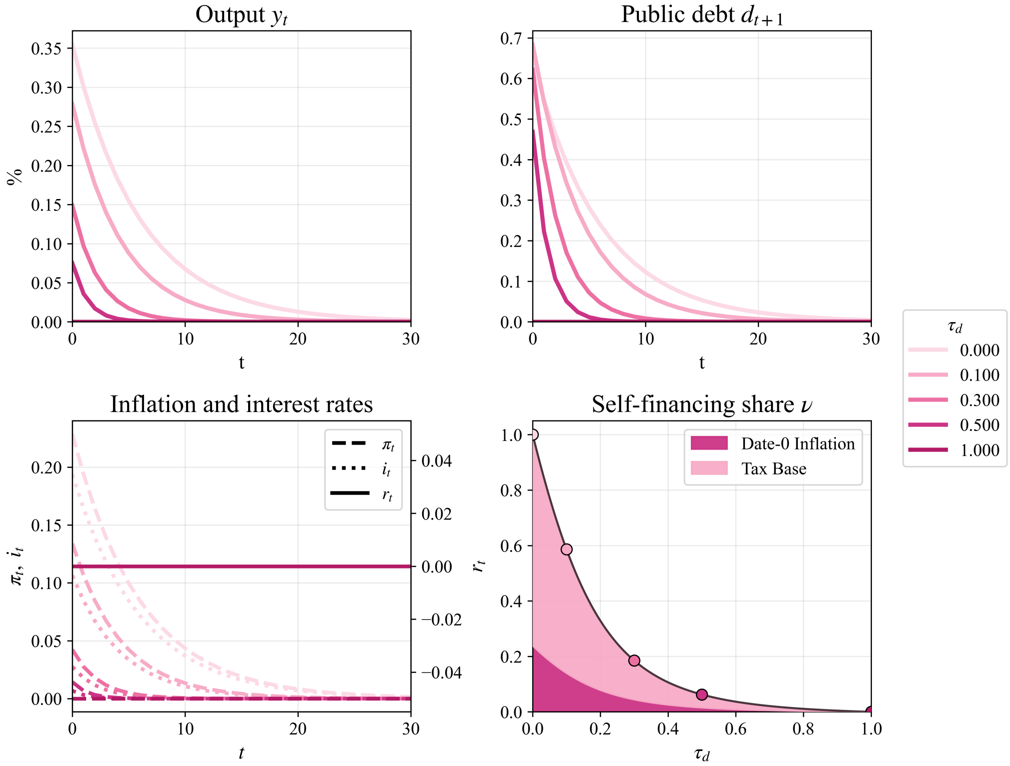

Figure 1: Impulse response functions for a fiscal deficit shock

Note: The first three panels show IRFs to a positive deficit shock of 1% of steady-state output Y^*. They are calculated by solving numerically for λ_s, χ and η in proposition 1. The vertical axes show percentage deviations from steady state, and colors indicate the fiscal adjustment strength τ_d. The final panel plots ν over $\tau_d \in (0, 1)$. The calibration follows table 1 with $\omega = 0.75$ and $\kappa = 0.1$.

Here the deficit shock raises demand because of non-Ricardian households ($\omega < 1$) and because of price rigidities, this higher demand translates into higher output. Lowering τ_d pushes repayments further into the future, making the transfer look larger to current households. Therefore both the output response, χ, and the persistence of the response, λ_s, increase. Thus, it raises, ν_y.

Inflation adds the second self-financing channel - when prices are not fully rigid (i.e. $\kappa > 0$). The boom then increases π_0 by $\eta(d_0 + e_0)$ and erodes the real value of outstanding debt. As we just argued, decreasing τ_d increases both χ and λ_s. Thus, it also increases the debt erosion channel, ν_π.

Together, this implies that, under $r_t = 0$, self-financing increases as fiscal adjustment is delayed, with $\nu \to 1$ as $\tau_d \to 0$, as shown in Angeletos et al. 2024.

Figure 2 shows that this logic changes under a hawkish monetary authority. First we note - not surprisingly - that the maximum self-financing share is lower for all $\tau_d \in (0, 1)$. The central bank responds hawkishly to both output and in-

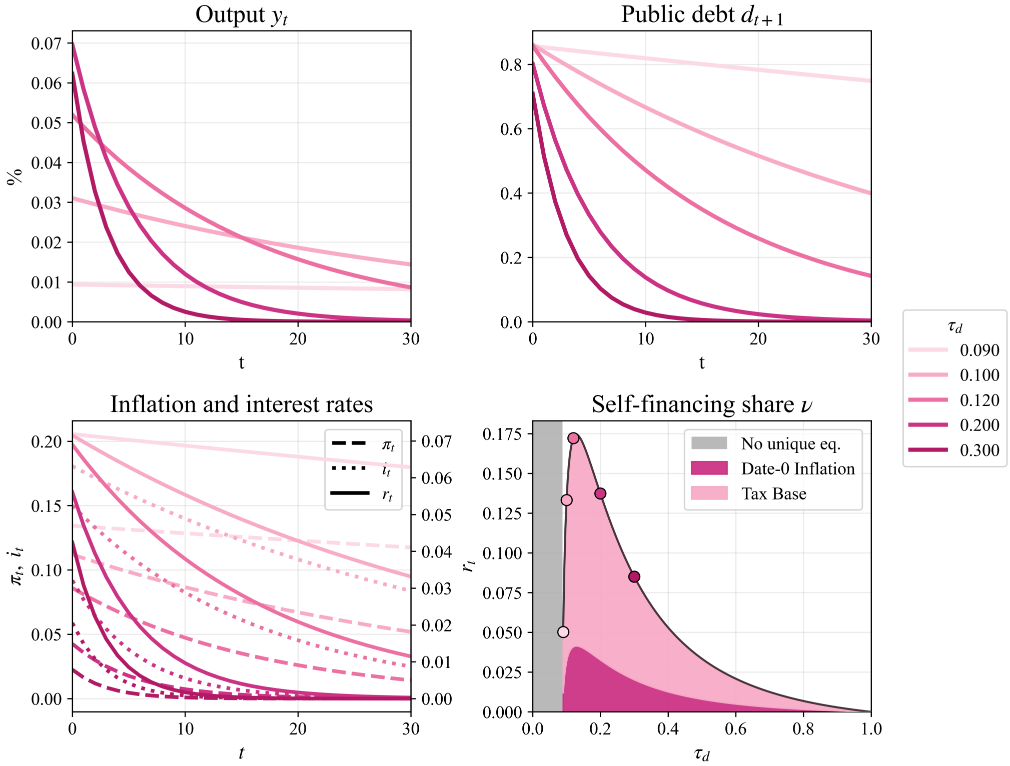

Figure 2: IRFs - Deficit shock under active monetary policy

Note: The plot is exactly as in Figure 1, except that monetary policy is hawkish with $\alpha_y = \alpha_\pi = 0.5$. The grey area in the bottom-right plot indicates parameterizations that do not satisfy the BK conditions and therefore do not lie in a stable Leeper region (Leeper 1991).

flation, which dampens demand when $X_r > 0$.[8] Higher real rates also increase government debt-servicing costs (i.e. the denominator of ν). Both effects work to reduce self-financing.

Perhaps more interesting, the relationship between τ_d and ν is changed and no longer monotone. This is because lowering τ_d now induces opposing forces. It still has the usual wealth effect: repayments are delayed, so current households perceive increased financial wealth which works to increase demand ($X_d > 0$).

However, reducing the fiscal adjustment rate, τ_d, directly increases the persistence of debt. This effect is amplified by higher real rates, which increase the roll-over costs of debt and thus slows repayment further (as discussed in 3.6). Higher real rates over a longer time period increases the debt servicing costs for the government and thus works to lower ν.

Inflation also increases as a result of higher persistence which again feeds back into higher rates. This new feedback mechanism can result in inflation increasing

[8] Recall that $X_r > 0$ when the intertemporal substitution effect of higher rates dominates the income effect from households owning government debt. This holds in the all calibrations of this paper.

while the output response χ is falling - as seen in figure 2.

The figure also shows that these negative forces - induced by the hawkish authority - can dominate, and thus, lowering τ_d reduces ν.

5 Quantitative analysis

I now turn to analyse the model quantitatively under empirically disciplined calibrations. Throughout, the economy is in steady state at $t = -1$, and the only shock is at $t = 0$, so that $\mathbb{E}_t[d_{t+k}] = d_{t+k}$ for $k > 0$.[9]

I solve the model numerically by imposing a truncation horizon T and impose $y_T = \pi_T = 0$. Following definition 1, I use aggregate demand (25), aggregate supply (29), the monetary rule (20) and the debt law of motion (17) to write the equilibrium conditions as a finite linear system,

$$H\boldsymbol{x} = \boldsymbol{h},$$

where

$$\boldsymbol{x} = [\pi_0, \pi_1, \dots, \pi_{T-1}, y_0, y_1, \dots, y_{T-1}, d_0, d_1, \dots, d_T]'.$$

The solution is then $\boldsymbol{x} = H^{-1}\boldsymbol{h}$. Since this procedure returns a path even in indeterminate cases, the code only solves parametrizations satisfying the BK conditions for a unique bounded equilibrium. (see G.3 and specifically the matrix (147))

5.1 Quantitative extension: HtM consumers

Like Angeletos et al. 2024 the analytical part of the paper has focussed on the highly tractable model with one type of household. In this quantitative section I, like Angeletos et al. 2024, expand the model with a fraction of credit-constrained consumers $\mu \in [0, 1)$[10]. This allows me to directly compare results with Angeletos et al. 2024, match empirical findings on intertemporal MPC and, as will be explained, to support the credibility of the analysis of anticipated transfer.

These *HtM* households hold monotone preference and cannot save. Therefore they consume all disposable income each period: $c_{i,t}^{HtM} = y_{i,t} - t_t$. Aggregate consumption is therefore

$$c_t = \mu c_t^{HtM} + (1-\mu) c_t^{OLG}, \tag{38}$$

where $c_t^{HtM} \equiv \int_0^1 c_{i,t}^{HtM}\, di$, and c_t^{OLG} is given by (24). The other change is that only

[9] The replication code is uploaded to a GitHub repository (link

[10] Also known as *hand-to-mouth* (HtM) consumers.

Parameter	Description	Value	Target	Value
Consumer spending				
μ	Share of HtM	0.073	Average *MPC* (See note)	0.2
ω	OLG survival rate	0.865	*MPC* slope (See note)	0.865
Policy				
τ_d	Tax feedback	.085	Galí, López-Salido, and Vallés 2007	
		.026	Bianchi and Melosi 2017	
		.004	Auclert and Rognlie 2020	
α_y	r_t feedback (output)	0	Angeletos et al. 2024	
α_π	r_t feedback (inflation)	0	Angeletos et al. 2024	
Nominal rigidities				
κ	NKPC slope	0.0062	Hazell et al. 2022	
Rest of the model				
σ	EIS	1	Standard (Woodford 2003)	
β	Discount factor	$0.99^{1/4}$	Annual real rate (ad hoc)	0.01
τ_y	Tax rate	0.33	Average labor tax (OECD) DeLong and Summers 2012	0.33
D^{ss}/Y^{ss}	Government debt	1.04	HH. liq. wealth Kaplan, Moll, and Violante 2018	1.04

Table 1: Model calibration

Note: The table shows the baseline model calibration. This is exactly the same baseline calibration Angeletos et al. 2024 consider in their quantitative section. The iMPC evidence comes from Fagereng, Holm, and Natvik 2021 that uses lottery winnings as their identification strategy. We use the impact $MPC_0 = 0.2$ value and the average MPC slope 0.865 and thus μ can be solved for.

OLG households hold debt (as private assets):

$$(1-\mu)\tilde{a}_t^{OLG} = d_t. \tag{39}$$

Combining (38) and (39) yields the hybrid aggregate consumption function (see G.1)

$$c_t = M_d d_t + M_y\Big((y_t - t_t) + \delta \mathbb{E}_t\Big[\sum_{k=1}^{\infty}(\beta\omega)^k(y_{t+k} - t_{t+k})\Big]\Big) - M_r \mathbb{E}_t\Big[\sum_{k=0}^{\infty}(\beta\omega)^k r_t\Big]. \tag{40}$$

Here $M_d \equiv 1 - \beta\omega$, $M_y = \mu + (1-\mu)(1-\beta\omega)$, $\delta \equiv \frac{(1-\mu)(1-\beta\omega)}{\mu+(1-\mu)(1-\beta\omega)}$, and $M_r \equiv (1-\mu)\xi$. Relative to the baseline model, the only substantive changes are that the aggregate MPC out of current disposable income is now larger than the MPC out of expected future disposable income (since $\delta < 1$ for $\mu > 0$) and that expected rates affect current consumption demand less. This carries over to the AD curve: expected transfers have a weaker direct effect on demand than a current transfer.

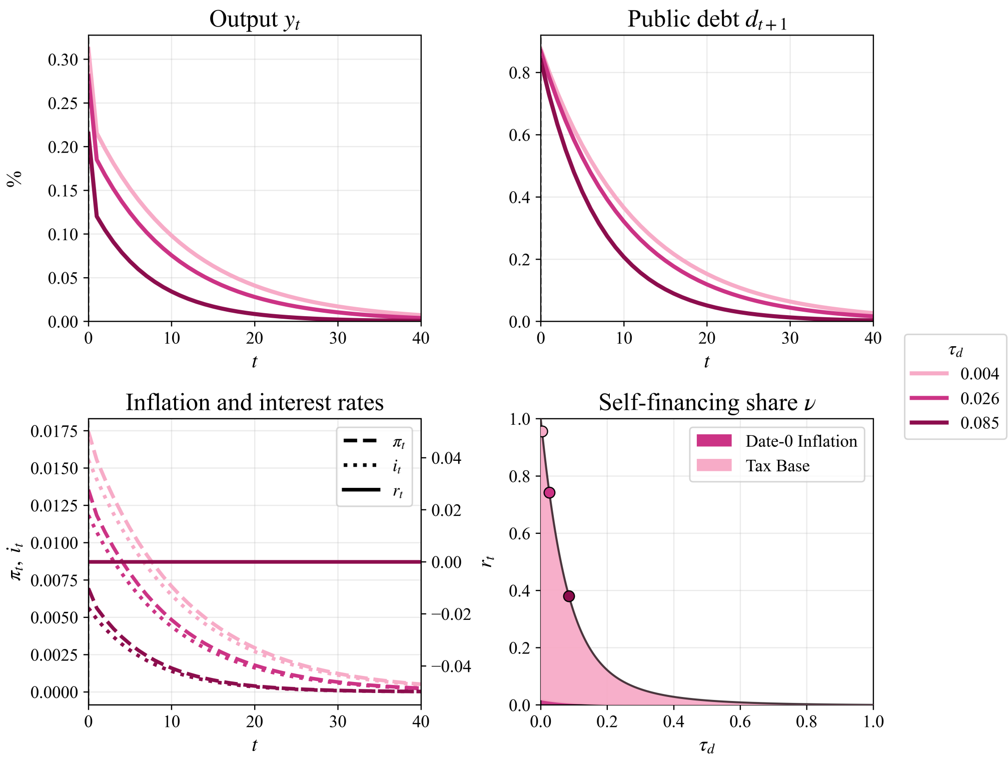

Figure 3: Impulse response functions for a fiscal deficit shock

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.

5.2 Neutral monetary policy

Table 1 presents the baseline calibration, which mirrors Angeletos et al. 2024. Each period t corresponds to a quarter.

Figure 3 replicates the quantitative results of Angeletos et al. 2024, corresponding to their Figure 3. The interpretation is equivalent to Figure 1 and we see that lower fiscal adjustment raises self-financing, with $\nu \to 1$ as $\tau_d \to 0^+$. For the estimated adjustment rates, self-financing is quantitatively large: $\nu_{0.004} \approx 0.95$, while $\nu_{0.085} \approx 0.38$ is still substantial.

With the empirically disciplined flat NKPC, $\kappa = 0.0062$, almost all self-financing comes from the tax-base channel rather than from the inflation channel.

The effect of *HtM* consumers is mainly at impact: the iMPC is higher in the period when the transfer is paid. Quantitatively, however, the effect is limited because the calibrated share $\mu = 0.073$ is small.[11]

[11] See H.1 for the same experiments in this section with $\mu = 0$.

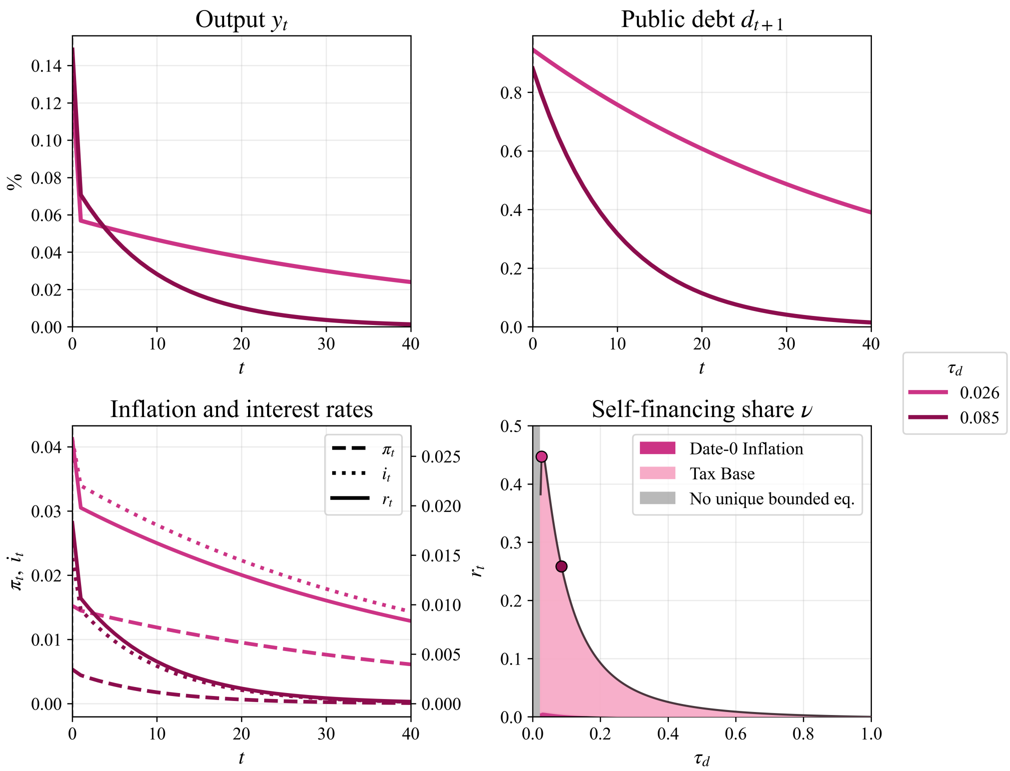

Figure 4: IRFs for a fiscal deficit shock under active monetary policy

Note: The first three panels show IRFs of a deficit shock of 1 pct. of SS output under active monetary policy. The final panel plots ν over $\tau_d \in (0, 1)$.

5.3 Active monetary policy

I now introduce active monetary policy and discipline the monetary rule using Smets and Wouters 2007.[12]

The resulting real-rate rule is

$$r_t = 0.0862 y_t + 1.0275 \pi_t. \tag{41}$$

Note that the exact real-rate rule calibration does not change the results substantially (see H.2 for the calibration $r_t = (0.5 + \kappa/\beta)y_t + (1.5 - 1/\beta)\pi_t$).

The results are shown in Figure 4. The main difference from the neutral-rate

[12] Smets and Wouters 2007 estimate the nominal interest-rate rule

$$i_t = \rho_i i_{t-1} + (1 - \rho_i)(r_\pi \pi_t + r_y y_t) + r_{\Delta y}(y_t - y_{t-1}) + \varepsilon_t^i,$$

with estimates $\hat{\rho}_i = 0.81$, $\hat{r}_\pi = 2.03$, $\hat{r}_y = 0.08$, $\hat{r}_{\Delta y} = 0.22$. By the equilibrium path real rate (30) this maps to

$$r_t = \left(0.08 + \frac{\kappa}{\beta}\right) y_t + \left(2.03 - \frac{1}{\beta}\right) \pi_t = 0.0862 y_t + 1.0275 \pi_t$$

benchmark is that the output and inflation responses now feed back into real rates. By Proposition 1, this affects the equilibrium through two channels. First, higher real rates dampen aggregate demand directly (in early periods), because households postpone consumption through intertemporal substitution. Second, higher real rates raise the cost of rolling over outstanding public debt, which increases debt-service costs and weakens self-financing.

This explains why active monetary policy lowers the self-financing share. The maximum output gap is roughly half that of $r_t = 0$, but more persistent. This mirrors the intuition of the fixed point relation (34). The maximal ν is therefore much lower than the possible $\nu_{max} = 1$ for $r_t = 0$, here about $\nu_{max} = 0.44$, with $\tau_d = 0.026$ close to the maximum.

As explained in the discussion of Figure 2 the role of fiscal delay is also different under active monetary policy. This explanation carries over to this hybrid model: lowering τ_d clearly increases persistence more relative to the neutral regime. This, together with a flatter output boom can break monotonicity.

5.4 Announced transfers in a neutral regime

The thesis so far has studied an unanticipated transfer (i.e. a deficit shock). I now consider an announced transfer: at time t, the government announces a debt-financed transfer to be implemented at $t+s$, $s \geq 0$.

This requires extending definition 2 to allow for expected transfers. Iterating forward the debt transition equation as in (35), I define the self-financing share as the ratio of self-financing sources to the total fiscal cost.

Definition 3. *The self-financing share of a transfer announced at t and carried out at time $t+s$, $s \geq 0$, is*

$$\nu_s \equiv \frac{\tau_y \sum_{k=0}^{\infty} \beta^k \mathbb{E}_0[y_k] + \frac{D^*}{Y^*}(\pi_0 - \mathbb{E}_{-1}[\pi_0])}{\sum_{k=0}^{\infty} \beta^k \mathbb{E}_0[e_k] + \frac{D^*}{Y^*} \sum_{k=0}^{\infty} \beta^{k+1} \mathbb{E}_0[r_k]}. \tag{42}$$

I first consider the case of neutral monetary policy with the calibration of table 1. Figure 5 shows the response to a transfer announced at $t = 0$ and implemented at $t = 20$.

Relative to the immediate transfer in Figure 3, the key difference is that the economy now responds before the transfer is paid. Forward-looking *OLG* households partly value the future transfer already at the announcement date. The announcement therefore creates a positive, pre-transfer output boom. This pre-transfer boom is central and leads to distinctly different results than the preceding quantitative sections.

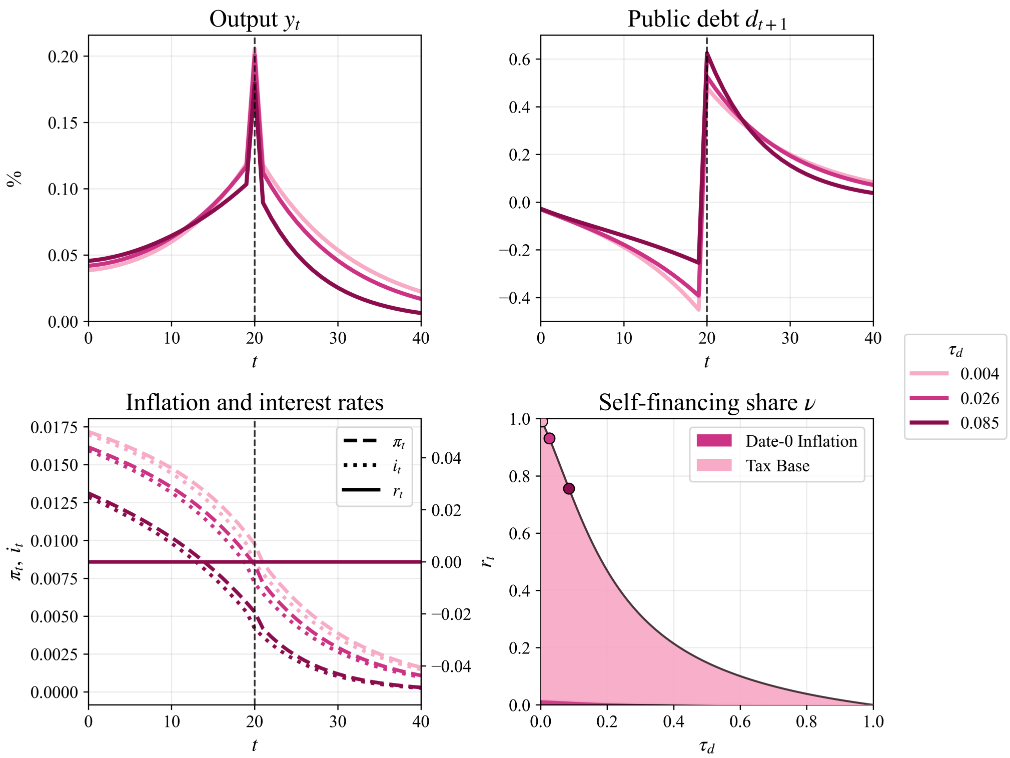

Figure 5: Announced transfer under neutral monetary policy

Note: The model is calibrated as in Table 1, with $r_t = 0$.

The pre-announcement boom materialises as lower debt before the transfer at $t = 20$ because of increased tax revenue. Hence the output boom before the transfer is caused by households using their financial assets to smooth consumption. With $\tau_d = 0.026$, for example, debt before implementation is approximately $d_{20} \approx -0.4$ percent of steady-state output. When the transfer is then paid, the debt is therefore lower - compared to the immediate transfer. In this sense, the transfer is partly pre-financed before it is implemented.[13]

The output path therefore has two parts. Delay creates a pre-transfer boom working to increase ν_{20}. But because this pre-transfer boom lowers inherited debt, the post-transfer boom starts from a smaller fiscal state than in the immediate-transfer experiment - therefore dampening the subsequent output boom.

This shift in timing of payments directly affects the self-financing share and comparing the lower-right panel of Figure 5 with Figure 3 reveals this: delaying implementation substantially increases self-financing. Most notably, for $\tau_d = 0.085$, the self-financing share rises from about $\nu_0 = 0.38$ for an immediate transfer to about $\nu_{20} = 0.76$ when implementation is delayed by 20 quarters.

[13] The strength of this pre-financing depends on τ_d. Lower τ_d means that fiscal adjustment is slower, so debt reductions generated by pre-transfer tax revenues are corrected more slowly. Debt is therefore further below steady state at implementation (see figure 5)

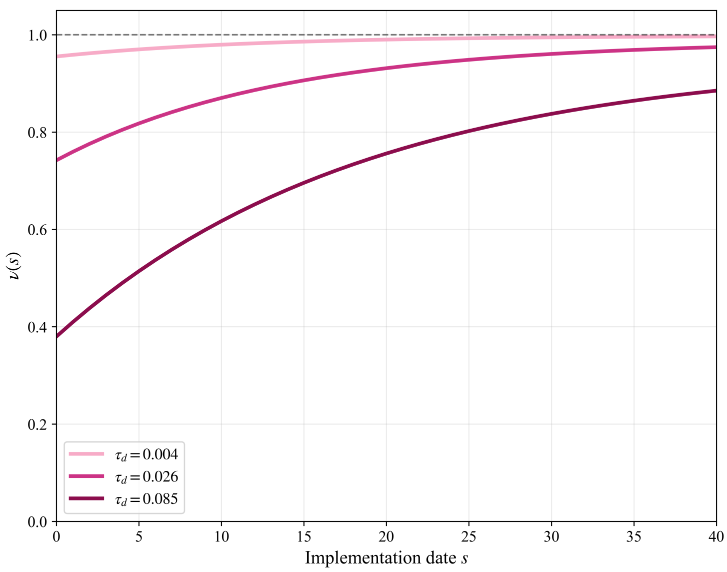

Figure 6: Self-financing by implementation delay under neutral monetary policy

Note: The figure plots ν_s for different implementation dates s, holding the announcement date fixed at $t = 0$. The model calibration follows Table 1. Parameter values that do not satisfy the BK condition are excluded.

Figure 6 shows the same mechanism for different implementation dates. We see that ν_s increases in s. A longer delay gives the pre-transfer tax-base channel more time to operate. More tax revenue is collected before the transfer is paid - where the PV of revenue is high. However, inherited debt is lower at implementation, and the remaining post-transfer fiscal state is smaller. Although this smaller post-transfer state reduces the post-transfer boom, the present-value gain from earlier tax revenues and the lower present value of the transfer cost dominate under neutral monetary policy. Since $r_t = 0$, there is no offsetting debt-service-cost channel. This is why implementation delay monotonically raises the self-financing share under neutral monetary policy.

5.5 Announced transfers in an active regime

Figure 7 repeats the announced-transfer experiment under the hawkish monetary policy (41). As in 5.3 $\tau_d = 0.004$ does not satisfy the BK condition and is omitted.

The main difference compared with the neutral announced case (i.e. figure 5) is that output initially falls below steady state. This happens even though the future transfer is perceived as private wealth (remember $\omega < 1$). The implementation is twenty quarters away, and therefore this direct demand effect is small. At the same time, the announced transfer implies a future boom. Since firms are forward looking, inflation rises already at announcement. Monetary policy responds

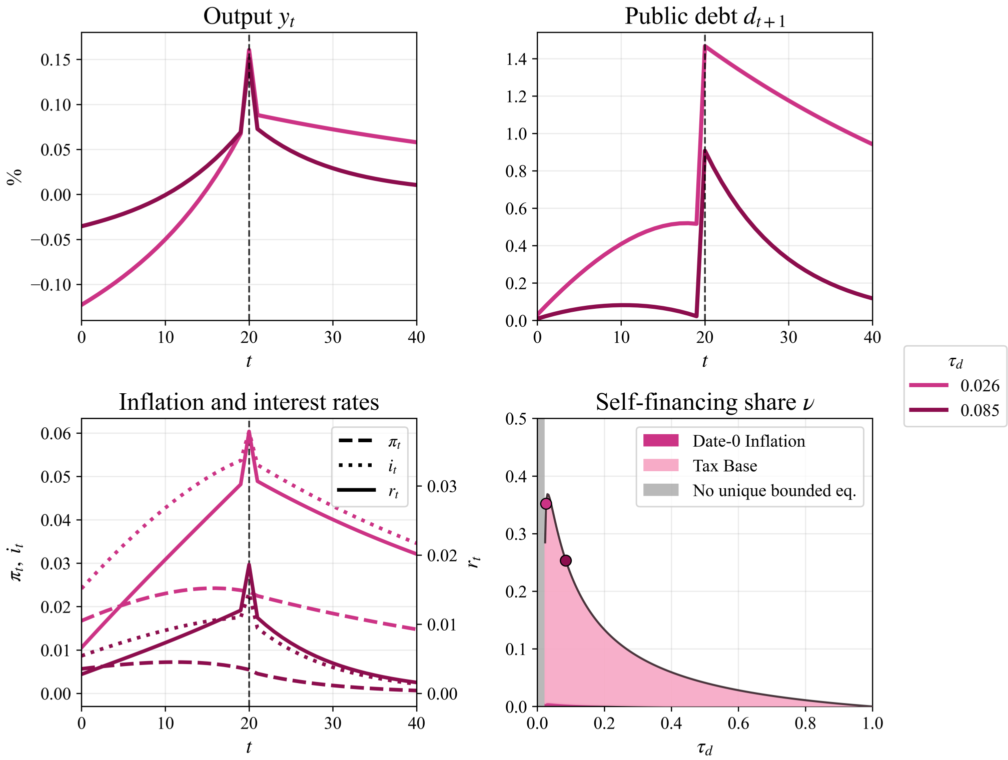

Figure 7: Announced transfer under active monetary policy

Note: The model calibration follows Table 1, except that $r_t = \alpha_y y_t + \alpha_\pi \pi_t = 0.086 y_t + 1.0275 \pi_t$. The transfer is announced at $t = 0$ and implemented at $s = 20$. Parameter values that do not satisfy the BK condition are excluded.

by raising the expected real-rate (starting at $t = 0$), and this contractionary real-rate effect dominates initially. Debt therefore initially rises because tax revenues fall while debt-servicing costs rise.

However, as $s = 20$ approaches, the perceived wealth of the transfer grows and output turns positive (which also reduces debt slightly). At implementation, the debt jumps because the transfer is paid. Because of the negative output gap, debt levels at $s = 20$ are higher compared to the neutral regime in figure 5. However, the positive effect of the prolonged output boom on ν_{20} is more than offset by the costly real rate and the pre-transfer negative output gap. This can be seen by comparing directly with the neutral regime in figure 5. For $\tau_d = 0.026$ the difference is most pronounced with $\nu_{20} \approx 0.93$ for the neutral case compared to the much lower $\nu_{20} \approx 0.36$. Thus, as with non-announced transfer, hawkish monetary policy suppresses output in early periods, increases the debt and output boom persistence and thus raises the debt servicing costs substantially. Together this results in ν_{20} being lower than ν_0 for all calibrations satisfying the BK conditions.

However, figure 8 shows that implementation delay - in our calibrations - can initially increase self-financing. As mentioned, this is because lower τ_d, ceteris

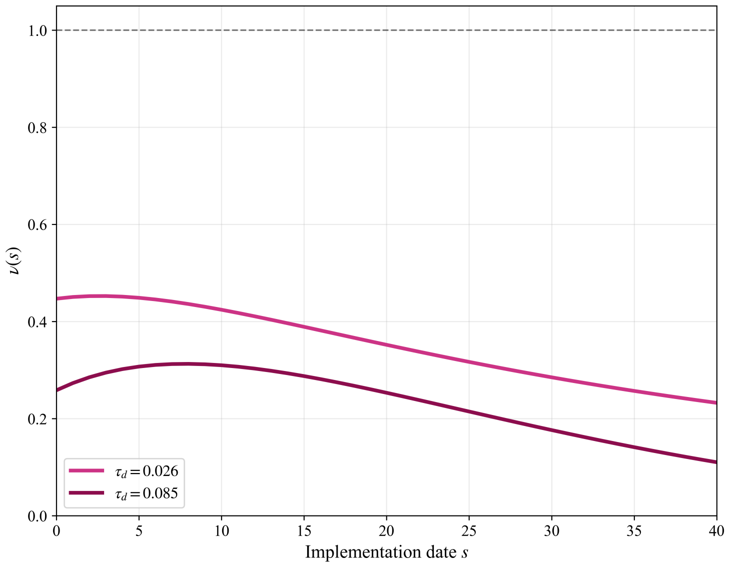

Figure 8: Self-financing by implementation delay under active monetary policy

Note: The figure plots ν_s for different implementation dates s, holding the announcement date fixed at $t = 0$. The model calibration follows Table 1, except that $r_t = \alpha_y y_t + \alpha_\pi \pi_t = 0.086y_t + 1.0275\pi_t$. Parameter values that do not satisfy the BK condition are excluded.

paribus, works to increase the pre-transfer output-boom. However, when implementation is delayed the negative effects win: lower output (even a negative gap) in the pre-transfer periods, increased persistence of debt, increased inflation and thus debt servicing costs.

For $\tau_d = 0.085$, figure 8 illustrates these competing effects clearly by the hump-shaped line. Self-financing rises from about 0.26 for an immediate transfer to about 0.31 around $s = 10$ (i.e. 10 quarters), before falling back to about 0.25 at $s = 20$. With faster fiscal adjustment, the debt state is less persistent, so a short delay allows some pre-transfer tax revenues to accrue without generating large servicing costs and early high inflation (depressing the demand). For longer delays, however, the accumulated real-rate costs dominate.

Taken together, this quantitative analysis highlights that monetary policy is central to the self-financing potential of fiscal deficits. Under neutral monetary policy, delayed fiscal adjustment and implementation delay both strengthen self-financing by allowing the demand expansion to raise output, broaden the tax base, and partly reduce the effective cost of the transfer before it is paid. Under active monetary policy, however, these mechanisms are weakened by higher real rates, lower output responses, and larger debt-service costs. The quantitative results therefore support the main theoretical point of the thesis: self-financing can be sizeable, but its strength is not automatic and depends crucially on the monetary regime.

6 Closing remarks

This thesis has studied how, and to what degree, deficit-financed stimulus checks can finance themselves, and how this depends on monetary policy and anticipation of the check. I do this in an environment with non-Ricardian perpetual-youth households and firms experiencing price rigidities. This allows fiscal deficits to increase output.

The main result is that active monetary policy can change self-financing both qualitatively and quantitatively, relative to the neutral policy. Under a neutral real rate, delayed fiscal adjustment raises self-financing, as in Angeletos et al. 2024. Under active monetary policy, this need not hold: higher output and inflation raise real rates, spread out demand, and increase debt-servicing costs. Thus, the same persistence that strengthens the tax-base channel also strengthens the servicing cost channel.

The quantitative analysis shows how monetary policy can be important for self-financing. Under the neutral regime, the model replicates large self-financing shares, mainly through the tax-base channel. Under active monetary policy, the maximum self-financing share is substantially lower. Announced transfers sharpen the point: with $r_t = 0$, implementation delay raises self-financing by generating tax revenues before the transfer is paid; with active monetary policy, delay instead creates a trade-off between potential pre-financing through the tax base and higher servicing costs.

These findings imply that self-financing is not a purely fiscal property. The same fiscal rule can imply very different outcomes depending on the monetary regime, especially when central banks respond strongly to inflation.

Interesting potential future work could study how self-financing changes in an open economy or with long-term nominal debt. An open economy would weaken the domestic tax-base channel if part of the demand boom falls on imports. Long-term bonds would make both inflation erosion and roll-over costs depend on the maturity structure. Introducing richer household heterogeneity could also be interesting. It could influence the strength of the demand boom depending on which households receive transfers and bear future taxes, while the inflation channel redistributes across households with different nominal asset positions.

References

Angeletos, George-Marios et al. (2024). "Can Deficits Finance Themselves?" In: Working paper.

Angeletos, George-Marios, Chen Lian, and Christian K. Wolf (2024). "Can Deficits Finance Themselves?" In: *Econometrica* 92.5, pp. 1351–1390. DOI: 10.3982/ECTA21791.

Auclert, Adrien and Matthew Rognlie (Jan. 2020). "Inequality and Aggregate Demand". Working paper. URL: https://web.stanford.edu/~aauclert/inequad.pdf.

Bianchi, Francesco and Leonardo Melosi (2017). "Escaping the Great Recession". In: *American Economic Review* 107.4, pp. 1030–1058. DOI: 10.1257/aer.20160186.

Blanchard, Olivier J. (1985). "Debt, Deficits, and Finite Horizons". In: *Journal of Political Economy* 93.2, pp. 223–247.

Blanchard, Olivier J. and Charles M. Kahn (1980). "The Solution of Linear Difference Models under Rational Expectations". In: *Econometrica* 48.5, pp. 1305–1311.

Calvo, Guillermo A. (1983). "Staggered Prices in a Utility-Maximizing Framework". In: *Journal of Monetary Economics* 12.3, pp. 383–398.

DeLong, J. Bradford and Lawrence H. Summers (2012). "Fiscal Policy in a Depressed Economy". In: *Brookings Papers on Economic Activity* 43.1, pp. 233–297. DOI: 10.1353/eca.2012.0000.

Dixit, Avinash K. and Joseph E. Stiglitz (1977). "Monopolistic competition and optimum product diversity". In: *American Economic Review* 67.3, pp. 297–308.

Fagereng, Andreas, Martin B. Holm, and Gisle J. Natvik (2021). "MPC Heterogeneity and Household Balance Sheets". In: *American Economic Journal: Macroeconomics* 13.4, pp. 1–54. DOI: 10.1257/mac.20190211.

Galí, Jordi (2015). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework*. 2nd ed. Princeton, NJ: Princeton University Press.

Galí, Jordi, J. David López-Salido, and Javier Vallés (2007). "Understanding the Effects of Government Spending on Consumption". In: *Journal of the European Economic Association* 5.1, pp. 227–270. DOI: 10.1162/JEEA.2007.5.1.227.

Hazell, Jonathon et al. (2022). "The Slope of the Phillips Curve: Evidence from U.S. States". In: *The Quarterly Journal of Economics* 137.3, pp. 1299–1344. DOI: 10.1093/qje/qjac010.

Kaplan, Greg, Benjamin Moll, and Giovanni L. Violante (2018). "Monetary Policy According to HANK". In: *American Economic Review* 108.3, pp. 697–743. DOI: 10.1257/aer.20160042.

Leeper, Eric M. (1991). "Equilibria under 'active' and 'passive' monetary and fiscal policies". In: *Journal of Monetary Economics* 27.1, pp. 129–147.

Smets, Frank and Rafael Wouters (2007). "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach". In: *American Economic Review* 97.3, pp. 586–606. DOI: 10.1257/aer.97.3.586.

Taylor, John B. (1993). "Discretion versus Policy Rules in Practice". In: *Carnegie-Rochester Conference Series on Public Policy* 39, pp. 195–214. DOI: 10.1016/0167-2231(93)90009-L.

Woodford, Michael (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press.

Yaari, Menahem E. (1965). "Uncertain Lifetime, Life Insurance, and the Theory of the Consumer". In: *Review of Economic Studies* 32.2, pp. 137–150.

7 Appendix

A Households

A.1 Consumption / savings optimisation

The household problem consists of the utility function (43)

$$\mathrm{U}_{i,t} \equiv \mathbb{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k (u(C_{i,t}) - v(L_{i,t})) \right], \tag{43}$$

where $u(C_{i,t}) \equiv \frac{C_{i,t+k}^{1-1/\sigma}-1}{1-\frac{1}{\sigma}}$ and $v(L_{i,t}) \equiv \iota \frac{L_{i,t+k}^{1+1/\varphi}}{1+\frac{1}{\varphi}}$ and the budget flow constraint (44)

$$A_{i,t+1} = \frac{I_t}{\omega}\Big(A_{i,t} + P_t(W_{i,t}L_{i,t} + Q_{i,t} - C_{i,t} - T_{i,t} + S_{i,t}) \Big). \tag{44}$$

Let $V_t(A_{i,t})$ be the maximum expected lifetime utility of a household at time t. This implies the following Bellman equation

$$V_t(A_{i,t}) = \max_{C_{i,t}} \Big\{ u(C_{i,t}) - v(L_{i,t}) + \beta\omega\mathbb{E}_t[V_{t+1}(A_{i,t+1})] \Big\} \quad \text{s.t.} \quad (44) \tag{45}$$

Assume that $V_t(A_{i,t})$ is differentiable and differentiate 45 wrt. $C_{i,t}$ - remember that $L_{i,t} = L_t$ is set by the labour union.

$$\begin{aligned} 0 &= \frac{\partial u}{\partial C_{i,t}} + \beta\omega\mathbb{E}_t \left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}} \frac{\partial A_{i,t+1}}{\partial C_{i,t}} \right] \\ &= \frac{\partial u}{\partial C_{i,t}} - \beta\omega\mathbb{E}_t \left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}} \frac{I_t}{\omega} P_t \right] \iff \\ \frac{\partial u}{\partial C_{i,t}} &= \beta I_t P_t \mathbb{E}_t \left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}} \right] \end{aligned} \tag{46}$$

We need an expression for $\partial V_{t+1}/\partial A_{i,t+1}$ which we will do explicitely, but one could also envelope the envelope theorem.

Define the function $F_t(A_{i,t})$ which optimises expected lifetime utility consumption given $A_{i,t}$ then (45) can be written as

$$V_t(A_{i,t}) = u(F_t(A_{i,t})) - v(L_{i,t}) + \beta\omega\mathbb{E}_t[V_{t+1}(A_{i,t+1})],$$

where $A_{i,t+1} = \frac{I_t}{\omega}\Big(A_{i,t} + P_t(Y_{i,t} - T_{i,t} - F(A_{i,t}) + S_{i,t}) \Big)$

Again remember that we assume differentiability of V_t so

$$\begin{aligned}\frac{\partial V_t}{\partial A_{i,t}} &= \frac{\partial}{\partial A_{i,t}}\Big(u(F_t(A_{i,t})) - v(L_{i,t}) + \beta\omega\mathbb{E}_t[V_{t+1}(A_{i,t+1})]\Big)\\ &= \frac{\partial u}{\partial F_t}\frac{\partial F_t}{\partial A_{i,t}} + \beta\omega\mathbb{E}_t\left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}}\frac{\partial A_{i,t+1}}{\partial A_{i.t}}\right]\end{aligned}\tag{47}$$

From the budget constraint (44) we obtain $\frac{\partial A_{t+1}}{\partial A_{i.t}} = \frac{I_t}{\omega}\left(1 - P_t\frac{\partial F}{\partial A_{i,t}}\right)$ inserting into (47) yields

$$\begin{aligned}\frac{\partial V_t}{\partial A_{i,t}} &= \frac{\partial u}{\partial F_t}\frac{\partial F_t}{\partial A_{i,t}} + \beta\omega\mathbb{E}_t\left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}}\frac{I_t}{\omega}\left(1 - P_t\frac{\partial F}{\partial A_{i,t}}\right)\right]\\ &= \frac{\partial u}{\partial F_t}\frac{\partial F_t}{\partial A_{i,t}} + \beta I_t\mathbb{E}_t\left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}}\right] - \beta I_t P_t\mathbb{E}_t\left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}}\right]\frac{\partial F}{\partial A_{i,t}}.\end{aligned}$$

By (46) and because $\partial u/\partial F \equiv \partial u/\partial C_{i,t}$ the first and last term cancel:

$$\begin{aligned}\frac{\partial V_t}{\partial A_{i,t}} &= \beta I_t\mathbb{E}_t\left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}}\right] \iff\\ P_t\frac{\partial V_t}{\partial A_{i,t}} &= \beta I_t P_t\mathbb{E}_t\left[\frac{\partial V_{t+1}}{\partial A_{i,t+1}}\right].\end{aligned}$$

The right hand side matches that of (46) and therefore

$$P_t\frac{\partial V_t}{\partial A_{i,t}} = \frac{\partial u}{\partial C_{i,t}} \iff P_{t+1}\frac{\partial V_{t+1}}{\partial A_{i,t+1}} = \frac{\partial u}{\partial C_{i,t+1}}$$

which inserting into (46) finally yields the general consumption Euler equation for the households:

$$\frac{\partial u}{\partial C_{i,t}} = \beta I_t P_t\mathbb{E}_t\left[\frac{1}{P_{t+1}}\frac{\partial u}{\partial C_{i,t+1}}\right] = \beta I_t\mathbb{E}_t\left[\frac{1}{\Pi_{t+1}}\frac{\partial u}{\partial C_{i,t+1}}\right]$$

For the specific CRRA utility we have

$$C_{i,t}^{-1/\sigma} = \beta I_t\mathbb{E}_t\left[\frac{C_{i,t+1}^{-1/\sigma}}{\Pi_{t+1}}\right].\tag{48}$$

A.2 Log linearizing the Euler

We utilize the following approximation when Log linearizing (5) around the deterministic zero inflation SS

$$C_{i,t}^{-1/\sigma} = C^*\exp\left\{-\frac{1}{\sigma}c_{i,t}\right\} \approx C^*\left(1 - \frac{c_{i,t}}{\sigma}\right), \quad I_t = \frac{1}{\beta}\exp\{i_t\} \approx \frac{1}{\beta}(1 + i_t) \text{ and}$$

$$\frac{1}{\Pi_t} = \frac{1}{\Pi^*}\exp\{-\pi_t\} \approx 1 - \pi_t.$$

Together (5) can be approximated by

$$C^{*-\frac{1}{\sigma}}\left(1 - \frac{c_{i,t}}{\sigma}\right) = \frac{\beta I^*}{\Pi^*}(1+i_t)\mathbb{E}_t\left[C^{*-\frac{1}{\sigma}}\left(1-\frac{c_{i,t+1}}{\sigma}\right)(1-\pi_{t+1})\right] \tag{49}$$

$$= C^{*-\frac{1}{\sigma}}(1+i_t)\mathbb{E}_t\left[\left(1-\frac{c_{i,t+1}}{\sigma}\right)(1-\pi_{t+1})\right] \tag{50}$$

$$= C^{*-\frac{1}{\sigma}}(1+i_t)\mathbb{E}_t\left[1-\frac{c_{i,t+1}}{\sigma} - \pi_{t+1} + \frac{c_{i,t+1}}{\sigma}\pi_{t+1}\right]. \tag{51}$$

Dropping second-order terms, $\frac{c_{i,t+1}}{\sigma}\pi_{t+1}$, $i_t\mathbb{E}_t[c_{i,t+1}]$, and $i_t\mathbb{E}_t[\pi_{t+1}]$, yields

$$C^{*-\frac{1}{\sigma}}\left(1 - \frac{c_{i,t}}{\sigma}\right) = C^{*-\frac{1}{\sigma}}\left(1 + i_t - \frac{1}{\sigma}\mathbb{E}_t[c_{i,t+1}] - \mathbb{E}_t[\pi_{t+1}]\right). \tag{52}$$

Cancelling $C^{*-\frac{1}{\sigma}}$ and rearranging then gives the log-linear Euler equation

$$c_{i,t} = \mathbb{E}_t[c_{i,t+1}] - \sigma\left(i_t - \mathbb{E}_t[\pi_{t+1}]\right). \tag{53}$$

where $c_{i,t} \equiv \ln(C_{i,t}) - \ln(C^*)$. Note that log linearizing $R_t \equiv I_t/\mathbb{E}_t[\Pi_{t+1}]$ yields:

$$\begin{aligned}
R^*(1+r_t) &= I^*(1+i_t)\mathbb{E}_t[(1-\pi_{t+1})] \iff \\
1 + r_t &= (1+i_t)\mathbb{E}_t[(1-\pi_{t+1}) \\
&= \mathbb{E}_t[1-\pi_{t+1}] + i_t\mathbb{E}_t[1-\pi_{t+1}] \\
&= 1 - \mathbb{E}_t[\pi_{t+1}] + i_t - i_t\mathbb{E}_t[\pi_{t+1}] \iff \\
r_r &= i_t - \mathbb{E}_t[\pi_{t+1}] - i_t\mathbb{E}_t[\pi_{t+1}]
\end{aligned}$$

dropping the second order term we have

$$r_t = i_t - \mathbb{E}_t[\pi_{t+1}] \tag{54}$$

A.3 Log linearizing the budget constraint

Throughout this part we denote old and newborns by indexing $h \in \{o, n\}$ meaning that $A^h_{i,j}$ is the assets for a newborn ($h = n$) or an old ($h = o$) agent in period t. Note that the budget constraint is conditional on surviving to period $t + 1$ where the people passing away simply get replaced by a new household with the same index and with $A^n_{i,t+1} = 0$. Let $h = n$ denote a newborn and $h = o$ denote an old household at period t. Using $W_{i,t} = W_t,\ L_{i,t} = L_t,\ Q_{i,t} = Q_t$ and $T_{i,t} = T_t$ we have

$$\begin{aligned}
A^h_{i,t+1} = \frac{I_t}{\omega}\left(A^h_{i,t} + P_t(W_tL_t + Q_t - C^h_{i,t} - T_t + S^h_{i,t})\right) \iff \\
\frac{A^h_{i,t+1}}{P_{t+1}} = \frac{I_t}{\omega\Pi_{t+1}}\left(\frac{\tilde{A}^h_{i,t}}{P_t} + Y_t - C^h_{i,t} - T_t\right) \iff \\
\frac{A^h_{i,t+1}}{P_{t+1}} + S^o_{t+1} = \frac{\tilde{A}^h_{i,t+1}}{P_{t+1}} = \frac{I_t}{\omega\Pi_{t+1}}\left(\frac{\tilde{A}^h_{i,t+1}}{P_t} + Y_t - C^h_{i,t} - T_t\right) + S^o
\end{aligned}$$

Remember that in steady state with market clearing private wealth is equal to public debt implying:

$$\begin{aligned}
A^* = \omega A^*_o + (1-\omega)A^*_n = \omega A^*_o = D^* \iff A^*_o = \frac{D^*}{\omega} \implies \\
\tilde{A}^*_o = \frac{D^*}{\omega} - \frac{1-\omega}{\omega}D^* = D^*
\end{aligned}$$

That is the steady state assets inclusive of the social fund is the same for both cohort (remember $S^n_t = S^n = D^*$. This in turn implies that consumption is equalised in steady state (since every household face the same utility function and budget constraint). Therefore the cohorts log linearised budget constraint (around the zero inflation SS) has the same structure as we will see.

By defining $r_t \equiv \ln(R_t) - \ln(R^*)$, $y_{i,t} \equiv \ln(Y_{i,t}) - \ln(Y^*)$ and $c^h_{i,t} \equiv \ln(C^h_{i,t}) - \ln(C^*)$ we have the following first order approximations:

$$\frac{I_t}{\Pi_{t+1}} \approx \frac{1}{\beta}(1 + i_t - \mathbb{E}_t[\pi_{t+1}]), \;\; Y_{i,t} \approx Y^*(1 + y_{i,t}), \;\; C^h_{i,t} \approx C^*(1 + c^h_{i,t})$$

and by defining $t_t \equiv (T_t - T^*)/Y^*$, $\tilde{a}^h_{i,t} = (\tilde{A}^h_{i,t}/P_t - D^*)/Y^*$. we have the identities

$$T_t = T^* + Y^*t_t, \;\; \frac{\tilde{A}^h_{i,t}}{P_t} = D^* + Y^*\tilde{a}^h_{i,t}$$

finally evaluating the government real values budget constraint in SS yields $D^* =$

$R^*(D^* - T^*)\mathbb{E}_t[\Pi^*]/\Pi^* \iff T^* = (1-\beta)D^*$ and we have that

$$\begin{aligned}
\frac{\tilde{A}^h_{i,t+1}}{P_{t+1}} &= \frac{I_t}{\omega\Pi_{t+1}}\left(\frac{\tilde{A}^h_{i,t+1}}{P_t} + Y_t - C^h_{i,t} - T_t\right) + S^o \\
&= \frac{I_t}{\omega\Pi_{t+1}}\Big(D^* + Y^*\tilde{a}^h_{i,t} + Y_t - C^h_{i,t} - (T^* + Y^* t_t)\Big) + S^o \\
&\approx \frac{1}{\beta\omega}(1 + i_t - \mathbb{E}_t[\pi_{t+1}])\Big(D^* - T^* + Y^*(\tilde{a}^h_{i,t} + (1+y_t) - (1+c^h_{i,t}) - t_t)\Big) + S^o \\
&\approx \frac{1}{\beta\omega}\Big(D^* - T^* + Y^*(\tilde{a}^h_{i,t} + (1+y_t) - (1+c^h_{i,t}) - t_t)\Big) \\
&\qquad\qquad + \frac{D^* - T^*}{\beta\omega}\Big(i_t - \mathbb{E}_t[\pi_{t+1}]\Big) + S^o \\
&= \frac{1}{\beta\omega}\Big(\beta D^* + Y^*(\tilde{a}^h_{i,t} + y_t - t_t - c^h_{i,t})\Big) + \frac{D^*}{\omega}\Big(i_t - \mathbb{E}_t[\pi_{t+1}]\Big) - \frac{1-\omega}{\omega}D^* \\
&= \frac{1 - 1 + \omega}{\omega}D^* + \frac{Y^*}{\beta\omega}\Big(\tilde{a}^h_{i,t} + y_t - t_t - c^h_{i,t}\Big) + \frac{D^*}{\omega}\Big(i_t - \mathbb{E}_t[\pi_{t+1}]\Big) \iff \\
\tilde{a}^h_{i,t+1} \equiv \frac{\frac{\tilde{A}^h_{i,t+1}}{P_{t+1}} - D^*}{Y^*} &= \frac{1}{\beta\omega}\Big(\tilde{a}^h_{i,t} + y_t - t_t - c^h_{i,t}\Big) + \frac{D^*}{\omega Y^*}\Big(i_t - \mathbb{E}_t[\pi_{t+1}]\Big) \qquad (55)
\end{aligned}$$

All in all we have the log linearized budget constraint for an economy without government spending given by (taking expectations at period t)

$$\mathbb{E}_t[\tilde{a}^h_{i,t+1}] = \frac{1}{\beta\omega}(\tilde{a}^h_{i,t} + (y_{i,t+k} - t_{t+k}) - c^h_{i,t}) + \frac{D^*}{\omega Y^*}(i_t - \mathbb{E}_t[\pi_{t+1}]) \qquad (56)$$

A.4 Linearized consumption function derivation

Rewriting we have

$$\begin{aligned}
\mathbb{E}_t[\tilde{a}^h_{i,t+1}] &= \frac{1}{\beta\omega}(\tilde{a}^h_{i,t} + (y_{i,t+k} - t_{t+k}) - c^h_{i,t}) + \frac{D^*}{\omega Y^*}(i_t - \mathbb{E}_t[\pi_{t+1}]) \iff \\
\tilde{a}^h_{i,t} &= \beta\omega\mathbb{E}_t[\tilde{a}^h_{i,t+1}] + c^h_{i,t} - (y_{i,t+k} - t_{t+k}) - \beta\frac{D^*}{Y^*}(i_t - \mathbb{E}_t[\pi_{t+1}])
\end{aligned}$$

Iterating forward (A.4) yields

$$\tilde{a}^h_{i,t} = \sum_{k=0}^{T}(\beta\omega)^k\left(c^h_{i,t+k} - (y_{i,t+k} - t_{t+k}) - \beta\frac{D^*}{Y^*}(i_{t+k} - \mathbb{E}_t[\pi_{t+k+1}])\right) + (\beta\omega)^{T+1}\mathbb{E}_t[\tilde{a}^h_{i,T+1}]$$

Imposing the transversality condition (57) ie. the no-ponzy condition for monotone utility in consumption.

$$\lim_{T\to\infty}\mathbb{E}_t[(\beta\omega)^{T+1}\tilde{a}^h_{i,t+T+1} = 0] \qquad (57)$$

yields the infinite recursive representation of the budget constraint:

$$\tilde{a}^h_{i,t} = \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k \left(c^h_{i,t+k} - (y_{i,t+k} - t_{t+k}) - \beta \frac{D^*}{Y^*} (i_{t+k} - \mathbb{E}_t[\pi_{t+k+1}]) \right) \iff \tag{58}$$

$$\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k \left((y_{i,t+k} - t_{t+k}) + \beta \frac{D^*}{Y^*} (i_{t+k} - \mathbb{E}_t[\pi_{t+k+1}]) \right) + \tilde{a}^h_{i,t} = \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (c^h_{i,t+k}) \tag{59}$$

In order to substitute $c^h_{t+k},\ k \geq 2$ for other state-variables (to solve the consumption function) we apply the Euler equation (**??**)

$$c^h_{i,t} = -\sigma(i_t - \pi_{t+1}) + \mathbb{E}_t[c^h_{i,t+1}] \iff \mathbb{E}_t[c^h_{i,t+1}] = \sigma(i_t - \pi_{t+1}) + c^h_{i,t} \implies$$
$$\mathbb{E}_t[c^h_{t+k}] = \sigma \sum_{j=0}^{k-1} (i_{t+j} - \pi_{t+j+1}) + c^h_t \tag{60}$$

Inserting (60) into (59) and reducing yields

$$\begin{aligned}
\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k \left((y_{i,t+k} - t_{t+k}) + \beta \frac{D^*}{Y^*} (i_{t+k} - \mathbb{E}_t[\pi_{t+k+1}]) \right) + \tilde{a}^h_{i,t}& \\
= \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (\sigma \sum_{j=0}^{k-1} (i_{t+j} - \pi_{t+j+1}) + c^h_{i,t})& \\
= \sigma \Big(\ (i_t - \pi_{t+1}) \ \ ((\beta\omega)^1 + (\beta\omega)^2 + (\beta\omega)^3 + (\beta\omega)^4 + \cdots)& \\
+ (i_{t+1} - \pi_{t+2})((\beta\omega)^2 + (\beta\omega)^3 + (\beta\omega)^4 + (\beta\omega)^5 + \cdots)& \\
+ (i_{t+2} - \pi_{t+3})((\beta\omega)^3 + (\beta\omega)^4 + (\beta\omega)^5 + (\beta\omega)^6 + \cdots) + \cdots \Big)& \\
= \frac{\sigma\beta\omega}{1-\beta\omega} \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (i_{t+k} - \pi_{t+k+1}) + \frac{c^h_{i,t}}{1-\beta\omega}&
\end{aligned}$$

Rearrange to isolate $c^h_{i,t}$.

$$\begin{aligned}
&\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k\Big((y_{i,t+k}-t_{t+k})+\beta\frac{D^*}{Y^*}(i_{t+k}-\mathbb{E}_t[\pi_{t+k+1}])\Big)+\tilde{a}^h_{i,t}\\
&\qquad=\frac{\sigma\beta\omega}{1-\beta\omega}\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(i_{t+k}-\mathbb{E}_t[\pi_{t+k+1}])+\frac{c^h_{i,t}}{1-\beta\omega}\iff\\
&(1-\beta\omega)\Big(\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(y_{i,t+k}-t_{t+k})+\mathbb{E}_t\sum_{k=0}^{\infty}\Big((\beta\omega)^k\beta\frac{D^*}{Y^*}(i_{t+k}-\mathbb{E}_t\pi_{t+k+1})\Big)+\tilde{a}^h_{i,t}\Big)\\
&\qquad=\sigma\beta\omega\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(i_{t+k}-\mathbb{E}_t[\pi_{t+k+1}])+c^h_{i,t}\iff\\
&c^h_{i,t}=(1-\beta\omega)\Big(\tilde{a}^h_{i,t}+\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(y_{i,t+k}-t_{t+k})\Big)\\
&\qquad+((1-\beta\omega)\beta\frac{D^*}{Y^*}-\sigma\beta\omega)\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(i_{t+k}-\mathbb{E}_t[\pi_{t+k+1}])
\end{aligned}$$

Define $\xi\equiv\sigma\beta\omega-(1-\beta\omega)\beta D^*/Y^*$ we arrive at the log linearised consumption function for household $i\in[0,1]$ (and $h\in\{o,n\}$).

$$c^h_{i,t}=(1-\beta\omega)\Big(\tilde{a}^h_{i,t}+\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(y_{i,t+k}-t_{t+k})\Big)-\xi\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(i_{t+k}-\mathbb{E}_t[\pi_{t+k+1}]) \tag{61}$$

Because of the social funds structure we have that (61) characterizes all consumers and therefore aggregating over $i\in[0,1]$ we obtain the aggregate log linearized consumption function or intertemporal Keynesian cross (IKC):

$$c_t=(1-\beta\omega)\Big(\tilde{a}_t+\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(y_{t+k}-t_{t+k})\Big)-\xi\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k(i_{t+k}-\mathbb{E}_t[\pi_{t+k+1}]) \tag{62}$$

where $c_t\equiv\int_0^1 c_{i,t}\,di$ and $\tilde{a}_t\equiv\int_0^1\tilde{a}_{i,t}\,di=d_tv$.

A.5 Union labour optimization

In each period t a worker union maximises the average (and total) utility of the households. Formally they solve

$$\max_{L_{i,t}}\int_0^1\Big(u(C_{i,t})-v(L_{i,t})\Big)\,di.\ \ s.t\ \ (44) \tag{63}$$

Letting $L_t = L_{i,t}$ and $W_t = W_{i,t}$ as described in the text. The unions problem (63) has the following FOC

$$\int_0^1 \left(u'(C_{i,t}) \frac{\partial C_{i,t}}{\partial L_t} - v'(L_{i,t}) \right) di = \int_0^1 \left(C_{i,t}^{-1/\sigma} (1-\tau_y) W_t - L_t^{1/\varphi} \right) di$$
$$= (1-\tau_y) W_t \int_0^1 C_{i,t}^{-1/\sigma} di - L_t^{1/\varphi} = 0$$

From here we isolate $(1-\tau_y)W_t$ and obtain the relation wage and the average marginal rate of substitution between consumption and labour

$$(1-\tau_y) W_t = \frac{L_t^{1/\varphi}}{\int_0^1 C_{i,t}^{-1/\sigma} di}. \tag{64}$$

Log linearizing By the assumptions consumption will be equal among individuals in steady state. Therefore we have

$$\int_0^1 C_{i,t}^{-1/\sigma} \, di = \int_0^1 (C^*)^{-1/\sigma} \exp\{-\frac{1}{\sigma} c_{i,t}\} \, di \approx (C^*)^{-1/\sigma} \int_0^1 1 - \frac{1}{\sigma} c_{i,t} \, di = (C^*)^{-1/\sigma} \left(1 - \frac{1}{\sigma} c_t \right)$$

Which together with (taking logs of (64))

$$\begin{aligned}
\ln\left(\frac{W_t}{W^*}\right) &= \frac{1}{\varphi} \ln\left(\frac{L_t}{L^*}\right) + \left\{ \ln\left(\int_0^1 C_{i,t}^{-1/\sigma} \right) di - \ln\left(\int_0^1 (C^*)^{-1/\sigma} \, di \right) \right\} \\
&\approx \frac{1}{\varphi} \ell_t + \ln\left((C^*)^{-1/\sigma} \left(1 - \frac{1}{\sigma} c_t \right) \right) - \ln\left((C^*)^{-1/\sigma} \right) \\
&= \frac{1}{\varphi} \ell_t + \ln\left(1 - \frac{1}{\sigma} c_t \right)
\end{aligned}$$

that is we approximately have

$$\frac{1}{\varphi} \ell_t = w_t - \frac{1}{\sigma} c_t \tag{65}$$

which is the log linearized condition for labour supply.

B Firms and aggregate supply

B.1 Final goods firm: Demand function

Final good firm takes prices as given and therefore optimizes the input-factor-basket according to the Dixit and Stiglitz (1977) aggregation technology:

$$\max_{Y_{j,t}} P_t Y_t - \int_0^1 P_{i,t} Y_{i,t} \, di \;\; s.t. \;\; Y_t = \left(\int_0^1 Y_{i,t}^{\frac{\varepsilon-1}{\varepsilon}} \, di \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

For a given intermediary good $j \in [0,1]$ the FOC is given by

$$
\begin{aligned}
P_t \frac{\varepsilon}{\varepsilon-1}\left(\int_0^1 Y_{i,t}^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}-1} \frac{\varepsilon-1}{\varepsilon} Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}-1} = P_{j,t} \iff \\
\left(\int_0^1 Y_{i,t}^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{1}{\varepsilon-1}} Y_{j,t}^{-\frac{1}{\varepsilon}} = \frac{P_{j,t}}{P_t} \iff \\
\left(\left(\int_0^1 Y_{i,t}^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}}\right)^{-1} Y_{j,t} = \left(\frac{P_{j,t}}{P_t}\right)^{-\varepsilon} \iff \\
Y_{j,t} = \left(\frac{P_{j,t}}{P_t}\right)^{-\varepsilon} Y_t,\ j \in [0,1] \qquad (66)
\end{aligned}
$$

Which is the final good firms demand function for firm j's good.

B.2 Final goods firm: Aggregate price index

Because the final good firm is perfectly competetive total revenue $(Y_t P_t)$ is equal to total costs $\int_0^1 P_{j,t} Y_{j,t}\, dj$ we can isolate P_t - the aggregate price index - using (66)

$$
\begin{aligned}
P_t Y_t &= \int_0^1 P_{j,t} Y_{j,t} dj = \int_0^1 P_{j,t}\left(\left(\frac{P_{j,t}}{P_t}\right)^{-\varepsilon} Y_t\right) dj \\
&= P_t^{\varepsilon} Y_t \int_0^1 P_{j,t} P_{j,t}^{-\varepsilon} dj = P_t^{\varepsilon} Y_t \int_0^1 P_{j,t}^{1-\varepsilon} dj
\end{aligned}
$$

isolating P_t we arrive at the aggregate priceindex

$$
P_t = \left(\int_0^1 P_{j,t}^{1-\varepsilon} dj\right)^{\frac{1}{1-\varepsilon}} \qquad (67)
$$

B.3 Intermediary good firms: Optimal reset price

Intermediary goods firm setting prices in period t maximises their expected future profits formally solving

$$
\max_{P^*_{j,t}} \sum_{k=0}^{\infty} E_t\left[Q^j_{t,t+k}\left(P^*_{j,t} \cdot Y_{j,t+k} - \Psi_{j,t+k}(Y_{j,t+k})\right)\right] \quad s.t\ \ Y_{j,t+k} = \left(\frac{P^*_{j,t}}{P_{t+k}}\right)^{-\epsilon} Y_{t+k}.
$$

where $\Psi(\cdot)$ denotes the cost function (in nominal terms), $Y_{t+k|t}$ denotes the real output of a firm that set its prices in period t and $Q_{t,t+k} \equiv \beta^k (C_{t+k}/C_t)^{-1/\sigma}(P_t/P_{t+k})$ is the stochastic discount factor.

To simplify we note that by the law of total probability we have (for each k)

$$
\begin{aligned}
&E_t\Big[Q^j_{t,t+k}\Big(P^*_{j,t}\cdot Y_{j,t+k}-\Psi_{j,t+k}(Y_{j,t+k})\Big)\Big] \\
&\qquad\qquad = \theta^k\mathbb{E}_t\Big[Q^j_{t,t+k}\Big(P^*_{j,t}\cdot Y_{j,t+k|t}-\Psi_{j,t+k}(Y_{j,t+k|t})\Big)\Big] \\
&\qquad\qquad + (1-\theta)\sum_{l=1}^{k}\theta^{k-l}\mathbb{E}_t\Big[Q^j_{t,t+k}\Big(P^*_{j,t+l}\cdot Y_{j,t+k|t+l}-\Psi_{j,t+k}(Y_{j,t+k|t+l})\Big)\Big]
\end{aligned}
$$

Since the second term does not depend on $P^*_{j,t}$ this drops out of the optimization problem and we are left with

$$
\max_{P^*_{j,t}}\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q^j_{t,t+k}\Big(P^*_tY_{j,t+k|t}-\Psi_{t+k}(Y_{j,t+k|t})\Big)\Big]\ \ s.t.\ \ Y_{j,t+k|t}=\left(\frac{P^*_{j,t}}{P_{t+k}}\right)^{-\epsilon}Y_{t+k}. \tag{68}
$$

That is: Firms setting their optimal price do so to maximise their expected discounted nominal profits for periods where they cannot change the price again.

First order condition By substituting $Y_{t+k|t}$ for the demand-constraints the problem simplifies to

$$
\begin{aligned}
&\max_{P^*_t}\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Bigg[Q_{t,t+k}\Bigg\{P^*_t\left(\frac{P^*_t}{P_{t+k}}\right)^{-\epsilon}Y_{t+k}-\Psi_{t+k}\Bigg(\left(\frac{P^*_t}{P_{t+k}}\right)^{-\epsilon}Y_{t+k}\Bigg)\Bigg\}\Bigg] \\
=&\max_{P^*_t}\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q_{t,t+k}\Big\{\big(P^*_t\big)^{1-\epsilon}P^{\epsilon}_{t+k}Y_{t+k}-\Psi_{t+k}\Big(\big(P^*_t\big)^{-\epsilon}P^{\epsilon}_{t+k}Y_{t+k}\Big)\Big\}\Big]
\end{aligned}
$$

The above problem have the following first order condition

$$
\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q_{t,t+k}\Big\{(1-\epsilon)\big(P^*_t\big)^{-\epsilon}P^{\epsilon}_{t+k}Y_{t+k}+\Big(\epsilon\big(P^*_t\big)^{-\epsilon-1}P^{\epsilon}_{t+k}Y_{t+k}\Big)\psi\Big(\big(P^*_t\big)^{-\epsilon}P^{\epsilon}_{t+k}Y_{t+k}\Big)\Big\}\Big]=0
$$

where $\psi=\Psi'$ is the nominal marginal cost function. Substituting $Y_{t+k|t}=\big(P^*_t\big)^{-\epsilon}P^{\epsilon}_{t+k}Y_{t+k}$ and then multiplying by $\big(P^*_t\big)/(1-\epsilon)$ can express the FOC as follows:

$$
\begin{aligned}
&\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q_{t,t+k}\Big\{(1-\epsilon)Y_{t+k|t}+\Big(\epsilon\big(P^*_t\big)^{-1}Y_{t+k|t}\Big)\psi\Big(Y_{t+k|t}\Big)\Big\}\Big]=0\iff \\
&\qquad\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q_{t,t+k}\Big\{P^*_tY_{t+k|t}+\Big(\frac{\epsilon}{1-\epsilon}\Big)Y_{t+k|t}\psi\Big(Y_{t+k|t}\Big)\Big\}\Big]=0
\end{aligned}
$$

Let M $\equiv\epsilon/(\epsilon-1)=-\epsilon/(1-\epsilon)$, $\psi_{t+k|t}\equiv\psi(Y_{t+k|t})$ and factorize out $Y_{t+k|t}$ to obtain the final representation of the first order condition in levels.

$$\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q_{t,t+k}Y_{t+k|t}\Big\{P_t^*-\mathrm{M}\psi_{t+k|t}\Big\}\Big]=0 \tag{69}$$

Log linearisation To explicitely solve for P_t^* we need to log linearize (69) around the zero inflation steady state. We do this around the zero inflation steady state. Therefore we first rewrite (69) in terms of variables well defined in the zero inflation SS. Specifically we divide by P_{t-1} and define $\Pi_{t-1,t+k}\equiv P_{t+k}/P_{t-1}$

$$\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q_{t,t+k}Y_{t+k|t}\Big\{\frac{P_t^*}{P_{t-1}}-\mathrm{M}MC_{t+k|t}\Pi_{t-1,t+k}\Big\}\Big]=0 \tag{70}$$

where we have defined the real marginal cost in period t+k for a firm that set prices in period t as $MC_{t+k|t}\equiv\psi_{t+k|t}/P_{t+k}$.

In steady state we have $\Pi^*=1$ (and $P^*=P_t=P_{t-1}=(P_t^*)^*$). Therefore at the zero-net-inflation steady state,

$$Q^*=\beta^k,\ \ (Y_{t+k|t})^*=Y^*,\ \ MC^*=\frac{1}{\mathrm{M}}.$$

Hence the factor in braces in (70) is zero in steady state ($1-\mathrm{M}\bar{MC}=0.$) and it is thus sufficient to log-linearize the factor in braces, evaluating $Q_{t,t+k}$ and $Y_{t+k|t}$ at steady state.[14]

Linearizing the factor in braces around the SS yields

$$\begin{aligned}\frac{P_t^*}{P_{t-1}}-\mathrm{M}MC_{t+k|t}\Pi_{t-1,t+k}&=\frac{P^*}{P^*}\exp\{p_t^*-p_{t-1}\}\\&\quad-\mathrm{M}MC^*\Pi^*\exp\Big\{\ln\Big(\frac{\mathrm{M}MC_{t+k|t}\Pi_{t-1,t+k}}{\mathrm{M}MC^*\Pi^*}\Big)\Big\}\\&=\exp\{p_t^*-p_{t-1}\}-\exp\{mc_{t+k|t}+\pi_{t-1,t+k}\}\\&\approx 1+p_t^*-p_{t-1}-(1+mc_{t+k|t}+\pi_{t-1,t+k})\\&=p_t^*-p_{t-1}-(mc_{t+k|t}+p_{t+k}-p_{t-1})\\&=p_t^*-(mc_{t+k|t}+p_{t+k}).\end{aligned} \tag{71}$$

where it is defined that $p_t^*\equiv\ln(P_t^*)-\ln(P^*)$, $\pi_{t-1,t+k}\equiv p_{t+k}-p_{t-1}$, $p_t\equiv\ln(P_t)-\ln(P^*)$, $mc_{t+k|t}\equiv\ln(MC_{t+k|t})-\ln(MC^*)$. In total - using (71) and (70) - the log linearised FOC for the reoptimizing intermediary goods firm $j\in[0,1]$ is

[14] This is valid to first order because $A\cdot B\approx A_0B_0+B_0(A-A_0)+A_0(B-B_0)=A_0B$ if $B_0=0$

given by

$$
\begin{aligned}
\sum_{k=0}^{\infty}\theta^k\mathbb{E}_t\Big[Q^*Y^*\{p_t^*-(mc_{t+k|t}+p_{t+k})\}\Big]=0 \iff\\
\sum_{k=0}^{\infty}(\beta\theta)^k\mathbb{E}_t\Big[\{p_t^*-(mc_{t+k|t}+p_{t+k})\}\Big]=0 \iff\\
p_t^*\sum_{k=0}^{\infty}(\beta\theta)^k=\sum_{k=0}^{\infty}(\beta\theta)^k\mathbb{E}_t\left[mc_{t+k|t}+p_{t+k}\right] \iff\\
p_t^*=(1-\beta\theta)\sum_{k=0}^{\infty}(\beta\theta)^k\mathbb{E}_t\left[mc_{t+k|t}+p_{t+k}\right]
\end{aligned}
\tag{72}
$$

From the optimal reset price to the NKPC This admits the recursive representation (derived just like the similar equations in section A.1 using $S_t^x \equiv \mathbb{E}_t\sum_{k=0}^{\infty}(\beta\theta)^k x_{t+k} = x_t+\beta\theta S_{t+1}^x$).

$$
\begin{aligned}
p_t^* &= (1-\beta\theta)(mc_{t|t}+p_t)+(1-\beta\theta)\left(\beta\theta\sum_{k=0}^{\infty}(\beta\theta)^k\mathbb{E}_t[mc_{t+k|t}+p_{t+k}]\right)\\
&=(1-\beta\theta)(mc_{t|t}+p_t)+\beta\theta\mathbb{E}_t[p_{t+1}^*]
\end{aligned}
\tag{73}
$$

Under linear technology and a common labour market, the real marginal cost is identical across firms, so $mc_{t|t}=mc_t$, where mc_t is the economy's average marginal cost in log deviations. Hence

$$
p_t^*=(1-\beta\theta)(mc_t+p_t)+\beta\theta\mathbb{E}_t[p_{t+1}^*]. \tag{74}
$$

Aggregate price index under Calvo pricing Because a fraction $1-\theta$ of firms reset their prices in period t, while a fraction θ keep last period's price, and using (67), the aggregate price index evolves as

$$
P_t^{1-\varepsilon}=\int_0^1 P_{j,t}^{1-\varepsilon}\,dj=\theta P_{t-1}^{1-\varepsilon}+(1-\theta)(P_t^*)^{1-\varepsilon}. \tag{75}
$$

Log linearizing (75) around the zero-inflation steady state yields

$$
\begin{aligned}
p_t=\theta p_{t-1}+(1-\theta)p_t^*. \iff\\
\pi_t=(1-\theta)(p_t^*-p_{t-1}) \iff\\
p_t^*-p_t=\frac{\theta}{1-\theta}\pi_t.
\end{aligned}
\tag{76}
$$

Deriving the NKPC in terms of real marginal cost Subtract p_t from both sides of (74):

$$\begin{aligned} p_t^* - p_t &= (1-\beta\theta) mc_t + \beta\theta \mathbb{E}_t[p_{t+1}^* - p_{t+1}] + \beta\theta \mathbb{E}_t[p_{t+1} - p_t] \\ &= (1-\beta\theta) mc_t + \beta\theta \mathbb{E}_t[p_{t+1}^* - p_{t+1}] + \beta\theta \mathbb{E}_t[\pi_{t+1}]. \end{aligned} \tag{77}$$

Now apply (76) both at time t and at time $t+1$:

$$p_t^* - p_t = \frac{\theta}{1-\theta}\pi_t, \qquad \mathbb{E}_t[p_{t+1}^* - p_{t+1}] = \frac{\theta}{1-\theta}\mathbb{E}_t[\pi_{t+1}].$$

Substituting into (77) gives

$$\begin{aligned} \frac{\theta}{1-\theta}\pi_t &= (1-\beta\theta) mc_t + \beta\theta \frac{\theta}{1-\theta}\mathbb{E}_t[\pi_{t+1}] + \beta\theta \mathbb{E}_t[\pi_{t+1}] \\ &= (1-\beta\theta) mc_t + \frac{\beta\theta}{1-\theta}\mathbb{E}_t[\pi_{t+1}]. & (78) \\ \pi_t &= \underbrace{\frac{(1-\theta)(1-\beta\theta)}{\theta}}_{\equiv \tilde{\kappa}} mc_t + \beta \mathbb{E}_t[\pi_{t+1}]. & (79) \end{aligned}$$

Linear technology and real marginal cost Now because the intermediary goods firms has a linear technology with exogenous steady technology $A > 0$,

$$Y_t = A N_t, \tag{80}$$

the real marginal cost is simply $MC_t = W_t/(P_t A)$. Since A is constant, log-deviations satisfy

$$mc_t = w_t, \tag{81}$$

Using the log linearized labour supply condition (65) market clearing $c_t = y_t$ and technology $y_t = \ell_t$ yields

$$\frac{1}{\varphi} y_t = w_t - \frac{1}{\sigma} y_t \iff w_t = \left(\frac{1}{\varphi} + \frac{1}{\sigma}\right) y_t. \tag{82}$$

Combining (81), (82) and (79) we the new Keynesian Phillips curve

$$\pi_t = \underbrace{\frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\frac{1}{\sigma} + \frac{1}{\varphi}\right)}_{\equiv \kappa} y_t + \beta \mathbb{E}_t[\pi_{t+1}]. \tag{83}$$

C Government

C.1 Fiscal authority

Nominal (flow) budget constraint Let B_t denote nominal public debt at the beginning of period t, and let T_t denote total real tax revenue. The nominal flow budget constraint of the fiscal authority is

$$\frac{1}{I_t} B_{t+1} = B_t - P_t T_t \tag{84}$$

Define real public debt as $D_t \equiv B_t / P_t$, gross inflation as $\Pi_{t+1} \equiv P_{t+1}/P_t$, and the ex-ante real interest rate as $R_t \equiv I_t / E_t[\Pi_{t+1}]$. Then (84) can be rewritten in real terms as

$$\frac{1}{I_t} B_{t+1} = B_t - P_t T_t \iff \tag{85}$$

$$\frac{B_{t+1}}{P_{t+1}} \equiv D_{t+1} = \frac{I_t}{P_t \Pi_{t+1}} (B_t - P_t T_t) = \frac{I_t}{E_t[\Pi_{t+1}]} (D_t - T_t) \frac{E_t[\Pi_{t+1}]}{\Pi_{t+1}}$$
$$= R_t (D_t - T_t) \left(\frac{E_t[\Pi_{t+1}]}{\Pi_{t+1}} \right). \tag{86}$$

This representation makes clear that unexpectedly high inflation between t and $t+1$ lowers the real value of outstanding nominal debt.

To allow for the possibility that steady-state debt is zero, define

$$d_t \equiv \frac{D_t - D^*}{Y^*}, \qquad t_t \equiv \frac{T_t - T^*}{Y^*}.$$

We now linearize (86) around the deterministic zero-inflation steady state. It is convenient to approximate each term separately. First,

$$R_t = R^* \exp(r_t) \approx R^*(1 + r_t) = \frac{1 + r_t}{\beta} \tag{87}$$

where we use $R^* = 1/\beta$. Second,

$$D_t - T_t = (D^* - T^*) + (D_t - D^*) - (T_t - T^*) = \beta D^* + Y^*(d_t - t_t), \tag{88}$$

where the last equality follows from the steady-state relation $T^* = (1 - \beta) D^*$. Finally,

$$\frac{E_t[\Pi_{t+1}]}{\Pi_{t+1}} = \frac{E_t[\exp(\pi_{t+1})]}{\exp(\pi_{t+1})} \approx \frac{E_t[1 + \pi_{t+1}]}{1 + \pi_{t+1}} \approx (1 + E_t[\pi_{t+1}])(1 - \pi_{t+1}) \approx 1 + E_t[\pi_{t+1}] - \pi_{t+1}. \tag{89}$$

Substituting (87), (88), and (89) into (86), and dropping second-order terms, yields

$$
\begin{aligned}
D_{t+1} &= R_t(D_t - T_t)\left(\frac{E_t[\Pi_{t+1}]}{\Pi_{t+1}}\right) \\
&\approx \frac{1+r_t}{\beta}(\beta D^* + Y^*(d_t - t_t))(1 + E_t[\pi_{t+1}] - \pi_{t+1}) \\
&\approx (1+r_t)\left(D^* + \frac{Y^*}{\beta}(d_t - t_t)\right)(1 + E_t[\pi_{t+1}] - \pi_{t+1}) \\
&\approx D^* + D^* r_t + \frac{Y^*}{\beta}(d_t - t_t) - D^*(\pi_{t+1} - E_t[\pi_{t+1}]).
\end{aligned}
$$

Hence the linearized law of motion for real public debt is

$$
d_{t+1} = \underbrace{\frac{D^*}{Y^*} r_t + \frac{1}{\beta}(d_t - t_t)}_{\text{expected debt next period}} - \underbrace{\frac{D^*}{Y^*}(\pi_{t+1} - E_t[\pi_{t+1}])}_{\text{erosion due to unexpected inflation}} . \tag{90}
$$

where $d_t \equiv (D_t - D^*)/Y^*$.

Tax policy The fiscal authority follows the baseline tax rule

$$
T_t = \tau_y Y_t + \bar{T} - \mathrm{E}_t + \tau_d(D_t - D^* + \mathrm{E}_t) \tag{91}
$$

where $\bar{T} \equiv T^* - \tau_y Y^*$, $\tau_y \in [0,1)$ is the proportional income tax, $\tau_d \in (0,1)$ governs the speed of fiscal adjustment, and E_t is a mean-zero deficit shock. Defining $e_t \equiv \mathrm{E}_t/Y^*$, the linearized tax rule is

$$
t_t = \tau_y y_t + \tau_d(d_t + e_t) - e_t \tag{92}
$$

C.2 Monetary authority

The monetary authority sets the nominal interest rate according to the policy rule

$$
\frac{I_t}{\Pi^*} = R^* \left(\frac{\Pi_t}{\Pi^*}\right)^{\psi} \left(\frac{Y_t}{Y^*}\right)^{\phi} \iff \frac{I_t}{I^*} = \left(\frac{\Pi_t}{\Pi^*}\right)^{\psi} \left(\frac{Y_t}{Y^*}\right)^{\phi} \tag{93}
$$

where $\psi, \phi \in \mathbb{R}$ govern the responsiveness of monetary policy to inflation and output respectively.

To linearize (93), define $i_t \equiv \ln(I_t) - \ln(I^*)$, $\pi_t \equiv \ln(\Pi_t) - \ln(\Pi^*)$, $y_t \equiv \ln(Y_t) - \ln(Y^*)$ and note that in the zero-inflation steady state we have $\Pi^* = 1$ and therefore $I^* = R^*$. Taking logs of (93) then yields

$$
i_t = \psi \pi_t + \phi y_t \tag{94}
$$

which is the log-linear nominal interest rate rule.

Using the definition of the ex-ante real interest rate,$R_t \equiv \frac{I_t}{E_t[\Pi_{t+1}]}$ we log linearize, $r_t = i_t - E_t[\pi_{t+1}]$ and substitute the monetary policy rule into this expression:

$$r_t = \psi\pi_t + \phi y_t - E_t[\pi_{t+1}] \tag{95}$$

Hence monetary policy can equivalently be represented as a rule for the real interest rate.

D Aggregate demand

D.1 Equation for discounted tax flows

Using the debt transition equation (90) and the fiscal policy (92) we have that

$$\begin{aligned}
\mathbb{E}_t[d_{t+k}] &= \mathbb{E}_t\left[\frac{D^*}{Y^*}r_{t+k-1} + \frac{1}{\beta}(d_{t+k-1} - t_{t+k-1}) - \frac{D^*}{Y^*}(\pi_{t+k} - \mathbb{E}_{t+k-1}[\pi_{t+k}])\right] \\
&= \mathbb{E}_t\left[\frac{D^*}{Y^*}r_{t+k-1} + \frac{1}{\beta}(d_{t+k-1} - t_{t+k-1})\right] \\
&= \mathbb{E}_t\left[\frac{D^*}{Y^*}r_{t+k-1} + \frac{1}{\beta}\Big(d_{t+k-1} - \{\tau_y y_{t+k-1} + \tau_d(d_{t+k-1} + e_{t+k-1}) - e_{t+k-1}\}\Big)\right] \\
&= \mathbb{E}_t\left[\frac{D^*}{Y^*}r_{t+k-1} + \frac{1}{\beta}\Big(d_{t+k-1}(1-\tau_d) - \tau_y y_{t+k-1} + e_{t+k-1}(1-\tau_d)\Big)\right] \qquad (96) \\
&= \begin{cases} d_t, & \text{for } k=0 \\ \mathbb{E}_t\left[\frac{D^*}{Y^*}r_t + \frac{1}{\beta}\Big(d_t(1-\tau_d) - \tau_y y_t\Big)\right] + \frac{1}{\beta}e_t(1-\tau_d), & \text{for } k=1 \\ \mathbb{E}_t\left[\frac{D^*}{Y^*}r_{t+k-1} + \frac{1}{\beta}\Big(d_{t+k-1}(1-\tau_d) - \tau_y y_{t+k-1}\Big)\right], & \text{for } k=2,3,... \end{cases}
\end{aligned} \tag{97}$$

using (97) yields

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k d_{t+k}\right] &= d_t + \omega e_t(1-\tau_d)\\
&\quad + \mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k\left(\frac{D^*}{Y^*}r_{t+k-1}+\frac{1}{\beta}\Big(d_{t+k-1}(1-\tau_d)-\tau_y y_{t+k-1}\Big)\right)\right]\\
&= d_t + \omega e_t(1-\tau_d) + \omega(1-\tau_d)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k d_{t+k}\right]\\
&\quad - \omega\tau_y\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right] + \beta\omega\frac{D^*}{Y^*}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right] \iff\\
\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k d_{t+k}\right] &= \frac{d_t}{1-\omega(1-\tau_d)} - \frac{\omega\tau_y}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right]\\
&\quad + \frac{\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right] + e_t\frac{\omega(1-\tau_d)}{1-\omega(1-\tau_d)}
\end{aligned} \tag{98}$$

using (98) and again (92) we have

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k t_{t+k}\right] &= \tau_y\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right] + \tau_d\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k d_{t+k}\right] - (1-\tau_d)e_t\\
&= \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right]\left(\tau_y - \frac{\omega\tau_y\tau_d}{1-\omega(1-\tau_d)}\right) + d_t\frac{\tau_d}{1-\omega(1-\tau_d)}\\
&\quad + \frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right] + e_t\left(\frac{\tau_d\omega(1-\tau_d)}{1-\omega(1-\tau_d)} - (1-\tau_d)\right)\\
&= \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right]\left(\frac{\tau_y(1-\omega(1-\tau_d))-\omega\tau_y\tau_d}{1-\omega(1-\tau_d)}\right)\\
&\quad + d_t\frac{\tau_d}{1-\omega(1-\tau_d)} + e_t\left(\frac{\tau_d\omega(1-\tau_d)-(1-\tau_d)(1-\omega(1-\tau_d))}{1-\omega(1-\tau_d)}\right)\\
&\quad + \frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right]
\end{aligned} \tag{99}$$

using that $\tau_y(1-\omega(1-\tau_d)) - \omega\tau_y\tau_d = \tau_y - \omega\tau_y + \omega\tau_d\tau_y - \omega\tau_d\tau_y = (1-\omega)\tau_y$ and $\tau_d\omega(1-\tau_d) - (1-\tau_d)(1-\omega(1-\tau_d)) = (1-\tau_d)(\tau_d\omega - 1 + \omega(1-\tau_d)) = (1-\tau_d)(\omega - 1)$ we arrive at

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k t_{t+k}\right] &= \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right] +\\
&+ \frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right] + \frac{\tau_d}{1-\omega(1-\tau_d)}d_t - \frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}e_t
\end{aligned} \tag{100}$$

D.2 From consumption, tax-flows and market clearing to the IS-curve

Using the log linearized consumption function (61) and market clearing $c_t = y_t$ along with (100) - which we just derived - we will derive the IS-curve.

$$\begin{aligned}
y_t &= (1-\beta\omega)\Big(d_t + \mathbb{E}_t \sum_{k=0}^{\infty}(\beta\omega)^k(y_{t+k} - t_{t+k})\Big) - \xi\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k} \\
&= d_t(1-\beta\omega)\Big(1 - \frac{\tau_d}{1-\omega(1-\tau_d)}\Big) + e_t\frac{(1-\beta\omega)(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)} \\
&\quad + (1-\beta\omega)\left(1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
&\quad - \left(\xi + (1-\beta\omega)\frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right]
\end{aligned} \tag{101}$$

(101) is reduced by utilizing that

$$\begin{aligned}
1 - \frac{\tau_d}{1-\omega(1-\tau_d)} &= \frac{1-\omega(1-\tau_d)-\tau_d}{1-\omega(1-\tau_d)} = \frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)} \text{ and} \\
\xi + \frac{(1-\beta\omega)\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)} &= \sigma\beta\omega - (1-\beta\omega)\beta\frac{D^*}{Y^*}\frac{(1-\omega(1-\tau_d)) - \tau_d\omega}{1-\omega(1-\tau_d)} \\
&= \sigma\beta\omega - \frac{D^*}{Y^*}\frac{\beta(1-\beta\omega)(1-\omega)}{1-\omega(1-\tau_d)} \\
&= (1-\beta\omega)\left(\frac{\sigma\beta\omega}{(1-\beta\omega)} - \frac{D^*}{Y^*}\frac{\beta(1-\omega)}{1-\omega(1-\tau_d)}\right)
\end{aligned}$$

we arrive at the dynamic IS-curve of the model stated in (102)

$$y_t = \mathrm{X}_d(d_t + e_t) + \mathrm{X}_y\mathbb{E}_t\left[(1-\beta\omega)\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right] - \mathrm{X}_r\mathbb{E}_t\left[(1-\beta\omega)\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right] \tag{102}$$

where

$$\mathrm{X}_d \equiv \frac{(1-\beta\omega)(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}, \quad \mathrm{X}_y \equiv 1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)},$$
$$\mathrm{X}_r \equiv \frac{\sigma\beta\omega}{(1-\beta\omega)} - \frac{D^*}{Y^*}\frac{\beta(1-\omega)}{1-\omega(1-\tau_d)}.$$

D.3 PIH AD-curve (recursive)

Note that by setting $\omega = 1$ we arrive at the PIH counterpart of the DIS curve:

$$y_t = \mathbb{E}_t\left[(1-\beta)\sum_{k=0}^{\infty}\beta^k y_{t+k}\right] - \sigma\beta\mathbb{E}_t\left[\sum_{k=0}^{\infty}\beta^k r_{t+k}\right] \tag{103}$$

which can be written in the recursive form given in the text (**??**):

$$
\begin{aligned}
y_t &= \mathbb{E}_t\left[(1-\beta)\sum_{k=0}^{\infty}\beta^k y_{t+k}\right] - \sigma\beta\mathbb{E}_t\left[\sum_{k=0}^{\infty}\beta^k r_{t+k}\right] \\
&= (1-\beta)y_t - \sigma\beta r_t + \mathbb{E}_t\left[(1-\beta)\sum_{k=1}^{\infty}\beta^k y_{t+k}\right] - \sigma\beta\mathbb{E}_t\left[\sum_{k=1}^{\infty}\beta^k r_{t+k}\right] \\
&= (1-\beta)y_t - \sigma\beta r_t + \beta\mathbb{E}_t\left[(1-\beta)\sum_{k=0}^{\infty}\beta^k y_{t+k+1}\right] - \sigma\beta^2\mathbb{E}_t\left[(1-\beta)\sum_{k=0}^{\infty}\beta^k r_{t+k+1}\right] \iff \\
y_t &= -\sigma r_t + \underbrace{\mathbb{E}_t\left[(1-\beta)\sum_{k=0}^{\infty}\beta^k y_{t+k+1}\right] - \sigma\beta\mathbb{E}_t\left[\sum_{k=0}^{\infty}\beta^k r_{t+k+1}\right]}_{=\mathbb{E}_t[y_{t+1}]}
\end{aligned}
$$

E Equilibrium

E.1 Deriving the linear system of equations

First write (102) in recursive form letting

$$
S_t^x \equiv \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k x_{t+k}\right] = x_t + \beta\omega\mathbb{E}_t[S_{t+1}^x], \quad x \in \{y, r\},
$$

$$
\begin{aligned}
y_t - X_d(d_t + e_t) &= (1-\beta\omega)X_y S_t^y - (1-\beta\omega)X_r S_t^r \\
&= (1-\beta\omega)X_y(y_t + \beta\omega\mathbb{E}_t[S_{t+1}^y]) - (1-\beta\omega)X_r(r_t + \beta\omega\mathbb{E}_t[S_{t+1}^r]) \\
&= (1-\beta\omega)X_y y_t - (1-\beta\omega)X_r r_t \\
&\qquad + \beta\omega\Big((1-\beta\omega)X_y E_t[S_{t+1}^y] - (1-\beta\omega)X_r\mathbb{E}_t[S_{t+1}^r]\Big) \\
&= (1-\beta\omega)X_y y_t - (1-\beta\omega)X_r r_t + \beta\omega\mathbb{E}_t[y_{t+1} - X_d(d_{t+1} + e_{t+1})].
\end{aligned} \tag{104}
$$

Now to find expressions for r_t and $q_t \equiv d_t + e_t$ in terms of the state variables y_t, q_t, π_t. First turning to the nominal interest rule directly deriving the r_t expression.

$$
r_t = i_t - \mathbb{E}_t[\pi_{t+1}] = \psi\pi_t + \phi y_t - \mathbb{E}_t[\pi_{t+1}] \tag{105}
$$

From the NKPC we have $\mathbb{E}_t[\pi_{t+1}] = \beta^{-1}(\pi_t - \kappa y_t)$. Inserting into (105):

$$
r_t = \Big(\psi - \frac{1}{\beta}\Big)\pi_t + \Big(\phi + \frac{\kappa}{\beta}\Big)y_t \tag{106}
$$

Now turning to deriving the relevant expression for q_t. Restating (97) for $k = 1$ in terms of q_t:

$$
\mathbb{E}_t[q_{t+1}] = \frac{D^*}{Y^*}r_t + \frac{1}{\beta}\Big((1-\tau_d)q_t - \tau_y y_t\Big) \tag{107}
$$

inserting (106) into (107):

$$\mathbb{E}_t[q_{t+1}] = \left(\frac{1-\tau_d}{\beta}\right)q_t + \left(\frac{D^*}{Y^*}\left(\phi+\frac{\kappa}{\beta}\right)-\frac{\tau_y}{\beta}\right)y_t + \frac{D^*}{Y^*}\left(\psi-\frac{1}{\beta}\right)\pi_t \tag{108}$$

Inserting (106) and (108) into (104) yields

$$\begin{aligned}
y_t - X_d q_t &= (1-\beta\omega)X_y y_t + \beta\omega\mathbb{E}_t[y_{t+1}] \\
&\quad - (1-\beta\omega)X_r\left(\left(\psi-\frac{1}{\beta}\right)\pi_t + \left(\phi+\frac{\kappa}{\beta}\right)y_t\right) \\
&\quad - \beta\omega X_d\left(\left(\frac{1-\tau_d}{\beta}\right)q_t + \left(\frac{D^*}{Y^*}\left(\phi+\frac{\kappa}{\beta}\right)-\frac{\tau_y}{\beta}\right)y_t + \frac{D^*}{Y^*}\left(\psi-\frac{1}{\beta}\right)\pi_t\right) \iff \\
\mathbb{E}_t[y_{t+1}] &= \frac{1}{\beta\omega}\left(1-(1-\beta\omega)\left(X_y - X_r\left(\phi+\frac{\kappa}{\beta}\right)\right) + \beta\omega X_d\left(\left(\phi+\frac{\kappa}{\beta}\right)\frac{D^*}{Y^*}-\frac{\tau_y}{\beta}\right)\right)y_t \\
&\quad + \frac{1}{\beta\omega}\left(\beta\omega X_d\left(\psi-\frac{1}{\beta}\right)\frac{D^*}{Y^*} + (1-\beta\omega)X_r\left(\psi-\frac{1}{\beta}\right)\right)\pi_t \\
&\quad - \left(\frac{X_d(1-\omega(1-\tau_d))}{\beta\omega}\right)q_t
\end{aligned} \tag{109}$$

E.2 Analysing the RE system of equations

The model is represented by the rational expectation system of equation given by (83), (108) and (109)

$$\mathbb{E}_t\begin{bmatrix} q_{t+1} \\ y_{t+1} \\ \pi_{t+1}\end{bmatrix} = \underbrace{\begin{bmatrix} m_{q,q} & m_{q,y} & m_{q,\pi} \\ m_{y,q} & m_{y,y} & m_{y,\pi} \\ 0 & -\frac{\kappa}{\beta} & \frac{1}{\beta}\end{bmatrix}}_{\equiv A}\begin{bmatrix} q_t \\ y_t \\ \pi_t\end{bmatrix} \tag{110}$$

where

$$\begin{aligned}
m_{q,q} &\equiv \frac{1-\tau_d}{\beta},\ m_{q,y} \equiv \left(\phi+\frac{\kappa}{\beta}\right)\frac{D^*}{Y^*}-\frac{\tau_y}{\beta},\ m_{q,\pi} \equiv \left(\psi-\frac{1}{\beta}\right)\frac{D^*}{Y^*}, \\
m_{y,q} &\equiv -\frac{X_d(1-\omega(1-\tau_d))}{\beta\omega} = -\frac{(1-\beta\omega)(1-\tau_d)(1-\omega)}{\beta\omega}, \\
m_{y,y} &\equiv \frac{1}{\beta\omega}\left(1-(1-\beta\omega)\left(X_y - X_r\left(\phi+\frac{\kappa}{\beta}\right)\right) + \beta\omega X_d\left(\left(\phi+\frac{\kappa}{\beta}\right)\frac{D^*}{Y^*}-\frac{\tau_y}{\beta}\right)\right), \\
m_{y,\pi} &\equiv \left(\psi-\frac{1}{\beta}\right)\left(\frac{(1-\beta\omega)X_r + \beta\omega X_d\frac{D^*}{Y^*}}{\beta\omega}\right).
\end{aligned}$$

E.3 Proof of lemma 1

Lemma 1 (repeated for convenience) *The system in (31) has a unique bounded equilibrium path if and only if the eigenvalues of the transition matrix A fulfill*

$|\lambda_s| < 1 < |\lambda_{u1}|, |\lambda_{u2}|$ *(without algebraic multiplicity)*

Proof. Let $h_t \equiv (B_t/P_{t-1} - D^*)/Y^*$ then

$$
\begin{aligned}
d_t &\equiv \frac{B_t/P_t - D^*}{Y^*} = \frac{1}{Y^*}\frac{B_t}{P_{t-1}}\frac{P_{t-1}}{P_t} - \frac{D^*}{Y^*} \\
&= \frac{\frac{1}{\Pi_t}(B_t/P_{t-1} - D^*) + \frac{1}{\Pi_t}D^*}{Y^*} - \frac{D^*}{Y^*} = \frac{1}{\Pi_t}h_t + \frac{D^*}{Y^*}\left(\frac{1}{\Pi_t} - 1\right)
\end{aligned}
$$

A first order taylor approximation of d_t around the zero-inflation SS yields:

$$
\begin{aligned}
d_t &\approx \frac{1}{\Pi^*}h^* + \frac{D^*}{Y^*}\left(\frac{1}{\Pi^*} - 1\right) + (h_t - h^*)\left(\frac{1}{\Pi^*}\right) + (\Pi_t - \Pi^*)\left(\frac{-1}{(\Pi^*)^2}h^* + \frac{D^*}{Y^*}\frac{-1}{(\Pi^*)^2}\right) \\
&= (h_t - 0)\cdot 1 + 1\cdot(\Pi_t - 1)\left(-0 - \frac{D^*}{Y^*}\right) \\
&= h_t - \frac{D^*}{Y^*}(1 + \Pi_t) \\
&\approx h_t - \frac{D^*}{Y^*}\pi_t,
\end{aligned}
$$

where $\pi_t \equiv \ln(\Pi_t) - \ln(\Pi^*) = \ln(\Pi_t)$

$$
d_t \approx h_t(1 - \pi_t) + \frac{D^*}{Y^*}(1 - \pi_t - 1)
$$

dropping second order terms:

$$
d_t \approx h_t - \pi_t\frac{D^*}{Y^*} \tag{111}
$$

Let

$$
\boldsymbol{d}_t \equiv \begin{pmatrix} d_t \\ y_t \\ \pi_t \end{pmatrix}, \qquad \boldsymbol{h}_t \equiv \begin{pmatrix} h_t \\ y_t \\ \pi_t \end{pmatrix}.
$$

Since $h_t = d_t + \frac{D^*}{Y^*}\pi_t$ (rearrange equation (111)), we have

$$
\boldsymbol{y}_t = \Theta\boldsymbol{h_t}, \qquad \Theta \equiv \begin{pmatrix} 1 & 0 & \frac{D^*}{Y^*} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.
$$

$det(\Theta) = 1$ thus Θ is invertible. If the $\boldsymbol{d}_t$-system is

$$
\mathbb{E}_t[\boldsymbol{y}_{t+1}] = A\boldsymbol{y}_t,
$$

then the h_t-system is

$$
\mathbb{E}_t[\boldsymbol{h}_{t+1}] = \Theta A\Theta^{-1}\boldsymbol{h}
$$

Hence the homogeneous transition matrices are similar: $A^h \equiv \Theta A \Theta^{-1}$.

Therefore the two systems have the same eigenvalues.[15] Moreover, since $\hat{x}_t = \Theta x_t$ is an invertible linear transformation,

$$\{x_t\}_{t\geq 0} \text{ is bounded } \iff \{\hat{x}_t\}_{t\geq 0} \text{ is bounded.}$$

Thus bounded equilibria are in one-to-one correspondence. Since h_t is predetermined and (y_t, π_t) are jump variables, the BK conditions for the h_t-system are exactly the condition for a unique bounded equilibrium in the equivalent d_t-system. □

E.4 Proof of proposition 1

Proposition 1 (repeated for convenience) Assume that A in (31) has one stable eigenvalue and two unstable eigenvalues, i.e. $|\lambda_s| < 1 < |\lambda_{u1}|, |\lambda_{u2}|$. Then the unique bounded equilibrium path is given by

$$\mathbb{E}_t[d_{t+1}] = \lambda_s(d_t + e_t), \qquad y_t = \chi(d_t + e_t), \qquad \pi_t = \eta(d_t + e_t), \tag{112}$$

with

$$d_0 = -\frac{\eta D^*/Y^*}{1 + \eta D^*/Y^*} e_0, \tag{113}$$

and where λ_s, χ, and η solve the fixed-point system

$$\chi = X_d + \frac{1-\beta\omega}{1-\beta\omega\lambda_s}\left[X_y\chi - X_r\left(\left(\phi + \frac{\kappa}{\beta}\right)\chi + \left(\psi - \frac{1}{\beta}\right)\eta\right)\right],$$
$$\lambda_s = \frac{1-\tau_d}{\beta} - \frac{\tau_y}{\beta}\chi + \frac{D^*}{Y^*}\left[\left(\phi + \frac{\kappa}{\beta}\right)\chi + \left(\psi - \frac{1}{\beta}\right)\eta\right], \quad \eta = \frac{\kappa\chi}{1-\beta\lambda_s}.$$

Proof. By Lemma 1, the d_t- and h_t-representations have the same eigenvalues and bounded equilibrium paths. Since the h_t-system has one predetermined variable, h_t, and two jump variables, y_t and π_t, the assumed root configuration satisfies the BK conditions. Hence there is a unique bounded equilibrium, and its path must eliminate the two unstable components of A. Let $\boldsymbol{x}_t \equiv \begin{bmatrix} q_t & y_t & \pi_t \end{bmatrix}'$ and $\boldsymbol{z}_t \equiv \begin{bmatrix} z_{s,t} & z_{u1,t} & z_{u2,t} \end{bmatrix} = V^{-1}\boldsymbol{x}_t$ where V^{-1} is the inverse of the well known eigenvector matrix stemming from $AV = V\Lambda$ where $\Lambda \equiv \text{diag}(\lambda_s, \lambda_{u1}, \lambda_{u2})$[16] Then

$$\mathbb{E}_t[\boldsymbol{x}_{t+1}] = A\boldsymbol{x}_t \iff \mathbb{E}_t[\boldsymbol{z}_{t+1}] = V^{-1}A\boldsymbol{x}_t = \Lambda V^{-1}\boldsymbol{x}_t = \Lambda\boldsymbol{z}_t$$

[15] Which can also quickly be confirmed:

$$\det(\lambda I - A^h) = \det(\lambda I - \Theta A\Theta^{-1}) = \det(\lambda\Theta\Theta^{-1} - \Theta A\Theta^{-1}) = \det\big(\Theta(\lambda I - A)\Theta^{-1}\big) = \det(\lambda I - A).$$

[16] Note we assume that V is invertible.

for clarity (114) restates this verbosely

$$\mathbb{E}_t \begin{bmatrix} z_{s,t+1} \\ z_{u1,t+1} \\ z_{u2,t+1} \end{bmatrix} = \begin{bmatrix} \lambda_s & 0 & 0 \\ 0 & \lambda_{u1} & 0 \\ 0 & 0 & \lambda_{u2} \end{bmatrix} \left(\begin{bmatrix} | & | & | \\ \boldsymbol{v}_s & \boldsymbol{v}_{u1} & \boldsymbol{v}_{u2} \\ | & | & | \end{bmatrix} \right)^{-1} \begin{bmatrix} q_t \\ y_t \\ \pi_t \end{bmatrix} = \begin{bmatrix} \lambda_s z_{s,t} \\ \lambda_{u1} z_{u1,t} \\ \lambda_{u2} z_{u2,t} \end{bmatrix} \tag{114}$$

By assumption $|\lambda_{u1}|, |\lambda_{u2}| > 1$ implying that neither $\mathbb{E}_t[z_{u1,t+k}] = (\lambda_{u1})^k z_{u1,t}$ or $\mathbb{E}_t[z_{u2,t+k}] = (\lambda_{u2})^k z_{u2,t}$ can be bounded by any $c \in \mathbb{R}$ unless $z_{u1,t} = z_{u2,t} = 0$. Therefore the bounded path must satisfy

$$\boldsymbol{z}_t = \begin{bmatrix} z_{s,t} \\ 0 \\ 0 \end{bmatrix} \iff \boldsymbol{x}_t = V\boldsymbol{z}_t = \boldsymbol{v}_s z_{s,t} \tag{115}$$

(which is exists and is unique because of the assumed bounds of A's eigenvalues and Blanchard and Kahn 1980) where $\boldsymbol{v}_s$ is the eigenvalue associated with λ_s (see (114)). Defining the stable eigenvector as $v_s \equiv \begin{bmatrix} 1 & \chi & \eta \end{bmatrix}'$ and by applying (115) we see that $q_t = z_{s,t}$, $y_t = \chi z_{s,t}$, $\pi_t = \eta z_{s,t}$ and thus have the following.[17]

$$\mathbb{E}_t[\boldsymbol{x}_{t+1}] = V\boldsymbol{z}_t = \lambda_s z_{s,t} \boldsymbol{v}_s \implies \mathbb{E}_t[q_{t+1}] = \lambda_s q_t$$

we have now derived (32). Turning to (34) we note that $A\boldsymbol{v}_s = \lambda_s \boldsymbol{v}_s$ which is the matrix representation of the following system of equations

$$\begin{cases} m_{q,q} + m_{q,y}\chi + m_{q,\pi}\eta & = \lambda_s \\ m_{y,q} + m_{y,y}\chi + m_{y,\pi}\eta & = \lambda_s \chi \\ \quad -\frac{\kappa}{\beta}\chi \;+\; \frac{1}{\beta}\eta & = \lambda_s \eta \end{cases} \tag{116}$$

The first equation yields

$$\lambda_s = \frac{1-\tau_d}{\beta} - \frac{\tau_y}{\beta}\chi + \frac{D^*}{Y^*}\left[\left(\phi + \frac{\kappa}{\beta}\right)\chi + \left(\psi - \frac{1}{\beta}\right)\eta\right].$$

Using the NKPC we get

$$\pi_t = \mathbb{E}_t[\pi_{t+1}] + \beta y_t \iff \eta q_t = \lambda_s \eta q_t + \beta \chi q_t \iff \eta = \frac{\kappa\chi}{1-\beta\lambda_s} \tag{117}$$

[17] Note we have normalized the vector such that the first entry is 1. This is possible since eigenvectors are characterised only by the relation between its elements (the direction in $\mathbb{R}^n$)

Finally insert $y_t = \chi(d_t + e_t)$, $\pi_t = \eta(d_t + e_t)$ in into aggregate demand (25) yields

$$\chi = X_d + \frac{1-\beta\omega}{1-\beta\omega\lambda_s}\left[X_y\chi - X_r\left(\left(\phi + \frac{\kappa}{\beta}\right)\chi + \left(\psi - \frac{1}{\beta}\right)\eta\right)\right]. \tag{118}$$

We pin down d_0 using the real debt transition equation (90) and the assumption that the model is in SS in period $t = -1$. Then

$$d_0 = -\pi_0\frac{D^*}{Y^*} = -\eta(d_0 + e_0)\frac{D^*}{Y^*} \iff d_0 = -\frac{\eta\frac{D^*}{Y^*}}{1+\eta\frac{D^*}{Y^*}} \tag{119}$$

□

E.5 Proof of Proposition 2

Proof. By Proposition 1, we have

$$\chi = X_d + \frac{1-\beta\omega}{1-\beta\omega\lambda_s}\left[X_y\chi - X_r\left(\alpha_y\chi + \alpha_\pi\eta\right)\right], \qquad \eta = \frac{\kappa\chi}{1-\beta\lambda_s}.$$

Since $|\lambda_s| < 1$ and $0 < \beta, \omega < 1$, we have $1-\beta\lambda_s > 0$, $1-\beta\omega\lambda_s > 0$, and hence

$$\Lambda \equiv \frac{1-\beta\omega}{1-\beta\omega\lambda_s} \in (0,1).$$

Substituting the Phillips-curve relation for η into the aggregate-demand fixed point gives

$$\chi = X_d + \Lambda\left[X_y - X_r\left(\alpha_y + \alpha_\pi\frac{\kappa}{1-\beta\lambda_s}\right)\right]\chi.$$

Define $M \equiv \alpha_y + \alpha_\pi\frac{\kappa}{1-\beta\lambda_s}$. Because monetary policy is hawkish, $\alpha_y \geq 0$ and $\alpha_\pi \geq 0$. Since $\kappa > 0$ and $1-\beta\lambda_s > 0$, this implies $M \geq 0$. The fixed point can therefore be written as

$$\chi = \frac{X_d}{1-\Lambda X_y + \Lambda X_r M}.$$

It remains to sign the numerator and denominator. From the definition of X_d,

$$X_d = \frac{(1-\beta\omega)(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)} > 0,$$

where the strict inequality follows from $\omega < 1$ and $\tau_d \in (0,1)$. Furthermore,

$$X_y = 1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}.$$

Using the standing assumption $0 \leq \tau_y < 1$, together with $1-\omega(1-\tau_d) = 1-\omega+\omega\tau_d > 1-\omega$, we get $0 < X_y \leq 1$. Finally, since $X_r \geq 0$, $M \geq 0$, and $\Lambda \in (0,1)$,

$1-\Lambda X_y+\Lambda X_r M \geq 1-\Lambda X_y \geq 1-\Lambda > 0$. The numerator is strictly positive and the denominator is strictly positive. Hence $\chi > 0$. $\square$

F Self financing

F.1 Period 0 shock

This section derives the formulas behind the self-financing discussion of the main text. Starting from the debt transition equation (97) we isolate d_t and iterate forward.

$$
\begin{aligned}
d_{t+1} &= \frac{1}{\beta}(d_t+e_t-\tau_d(d_t+e_t)-\tau_y y_t)+\frac{D^*}{Y^*}r_t-\frac{D^*}{Y^*}(\pi_{t+1}-\mathbb{E}_t[\pi_{t+1}]) \iff \\
d_t &= \beta d_{t+1}-e_t+\tau_d(d_t+e_t)+\tau_y y_t-\beta\frac{D^*}{Y^*}r_t+\beta\frac{D^*}{Y^*}(\pi_{t+1}-\mathbb{E}_t[\pi_{t+1}]) \iff \\
d_t &= \beta\mathbb{E}_t[d_{t+1}]-e_t+\tau_d(d_t+e_t)+\tau_y y_t-\beta\frac{D^*}{Y^*}r_t \\
&= \beta^2\mathbb{E}_t[d_{t+2}]+\tau_d(e_t+\beta d_{t+1}+d_t)+\tau_t(\beta y_{t+1}+y_t)-\frac{D^*}{Y^*}\left(\beta^2\mathbb{E}_t[r_{t+1}]+\beta r_t\right)-e_t \\
&= \mathbb{E}_t[\beta^T d_{t+T}]+\tau_d\left(e_t+\sum_{k=0}^{T-1}\beta^k\mathbb{E}_t[d_{t+k}]\right)+\tau_y\left(\sum_{k=0}^{T-1}\beta^k\mathbb{E}_t[y_{t+k}]\right) \\
&\qquad -\frac{D^*}{Y^*}\left(\sum_{k=0}^{T-1}\beta^{k+1}\mathbb{E}_t[r_{t+k}]\right)-e_t
\end{aligned}
$$

In the limit $T\to\infty$ because of stability (see proposition E.4) $\lim_{T\to\infty}\mathbb{E}_t[\beta^T d_{t+T}]=0$ and we have

$$
\begin{aligned}
d_t &= \tau_d\left(e_t+\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t[d_{t+k}]\right)+\tau_y\left(\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t[y_{t+k}]\right)-\frac{D^*}{Y^*}\left(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_t[r_{t+k}]\right)-e_t \iff \\
e_t+\frac{D^*}{Y^*}\left(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_t[r_{t+k}]\right) &= \tau_d\left(e_t+\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t[d_{t+k}]\right)+\tau_y\left(\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t[y_{t+k}]\right)-d_t
\end{aligned}
$$

Using that with the economy in steady state in period $t=-1$ $d_0=-\pi_0 D^*/Y^*$ we arrive at the wanted representation of decomposing the initial deficit shock:

$$
\begin{aligned}
e_0+\underbrace{\frac{D^*}{Y^*}\left(\sum_{k=0}^{\infty}\beta^{k+1}\mathbb{E}_0[r_k]\right)}_{\text{Increased debt servecing costs}} &= \underbrace{\tau_d\left(e_0+\sum_{k=0}^{\infty}\beta^k\mathbb{E}_0[d_k]\right)}_{\text{Fiscal adjustment}} \\
&\quad + \underbrace{\tau_y\left(\sum_{k=0}^{\infty}\beta^k\mathbb{E}_0[y_k]\right)}_{\text{SF from taxbase expansion}} + \underbrace{\frac{D^*}{Y^*}(\pi_0-\mathbb{E}_{-1}[\pi_0])}_{\text{SF from period 0 inflation}}
\end{aligned} \tag{120}
$$

F.2 Proof of theorem 1

Theorem 1 (repeated for convenience) Assume A in (110) has eigenvalues that satisfy $|\lambda_s| < 1 < |\lambda_{u1}|, |\lambda_{u2}|$ then proposition 1 holds. Assume too that χ is differentiable in $\tau_d \in (0,1)$ then given a one off fiscal deficit shock ($e_0 \neq 0$ and $e_t = 0 \; \forall t \geq 1$) the degree of self financing ν defined as (36) satisfies

$$\frac{\partial \nu}{\partial \tau_d} < 0 \iff \tau_d \frac{\partial \chi}{\partial \tau_d} < \chi \tag{121}$$

Which does not generally hold. That is the self-financing share ν is not monotonically increasing with decreasing fiscal adjustment rate τ_d!

Proof. The RE system (110) has

$$r_t = \Big(\phi + \frac{\kappa}{\beta}\Big) y_t + \Big(\psi - \frac{1}{\beta}\Big)\pi_t = \alpha_y y_t + \alpha_\pi \pi_t$$

where $\alpha_y \equiv \phi + \kappa/\beta$ and $\alpha_\pi \equiv \psi - 1/\beta$. By proposition 1 this can be written in terms of q_t:

$$r_t = (\alpha_y \chi + \alpha_\pi \eta) q_t = \Xi q_t \tag{122}$$

where $\Xi \equiv \alpha_y \chi + \alpha_\pi \eta$. Using (122) and the first equation of the eigenvector condition $A v_s = \lambda_s \boldsymbol{v}_s$ yields

$$\begin{aligned}
\underbrace{\frac{1-\tau_d}{\beta}}_{m_{q,q}} + \Bigg(\underbrace{\frac{D^*}{Y^*}\alpha_y - \frac{\tau_y}{\beta}}_{m_{q,y}}\Bigg)\chi + \Bigg(\underbrace{\frac{D^*}{Y^*}\alpha_\pi}_{m_{q,\pi}}\Bigg)\eta = \lambda_s \iff \\
1 - \tau_d + \beta\frac{D^*}{Y^*}\Big(\alpha_y \chi + \alpha_\pi \eta\Big) - \beta\frac{\tau_y}{\beta}\chi = \beta\lambda_s \iff \\
1 - \beta\lambda_s = \tau_d + \tau_y \chi - \beta\frac{D^*}{Y^*}\Xi
\end{aligned} \tag{123}$$

Now we can calculate the self-financing ratio as a function of the systems parameters (and stable eigenvalue). Using that $\mathbb{E}_0[q_k] = \lambda_s^k q_0 \implies \mathbb{E}_0[y_t] = \chi\lambda_s^k q_0, \mathbb{E}_0[r_t] = \Xi\lambda_s^k q_0$ we have

$$\sum_{k=0}^{\infty} \beta^k \mathbb{E}_0[y_t] = \sum_{k=0}^{\infty} \beta^k \chi \lambda_s^k q_0 = \chi q_0 \sum_{k=0}^{\infty} (\beta\lambda_s)^k = \frac{\chi q_0}{1-\beta\lambda_s} \tag{124}$$

$$\sum_{k=0}^{\infty} \beta^{k+1} \mathbb{E}_0[r_t] = \sum_{k=0}^{\infty} \beta^{k+1} \Xi \lambda_s^k q_0 = \Xi q_0 \beta \sum_{k=0}^{\infty} (\beta\lambda_s)^k = \frac{\beta \Xi q_0}{1-\beta\lambda_s} \tag{125}$$

Inserting (124) and (125) into (36) and using that by proposition E.4 $q_0 = d_0 + e_0 =$

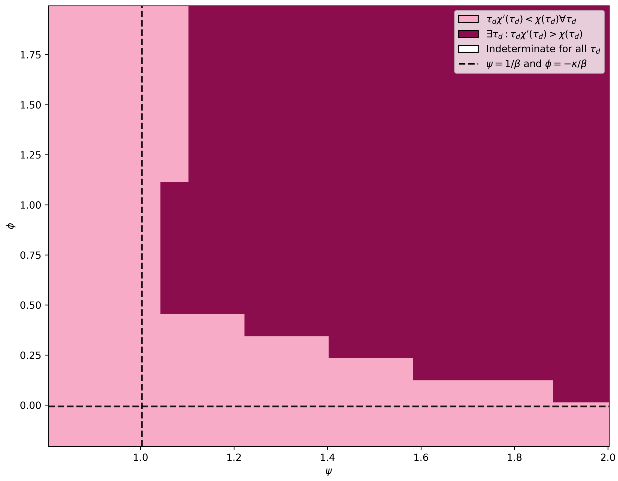

Figure 9: Numerical check of monotonicity condition

$e_0/(1+\eta D^*/Y^*) \iff e_0 = q_0(1+\eta D^*/Y^*)$ yields

$$\nu = \frac{\tau_y \frac{\chi q_0}{1-\beta\lambda_s} + \frac{D^*}{Y^*}\eta q_0}{e_0 + \frac{D^*}{Y^*}\frac{\beta\Xi q_0}{1-\beta\lambda_s}} = \frac{\tau_y \frac{\chi q_0}{1-\beta\lambda_s} + \frac{D^*}{Y^*}\eta q_0}{q_0(1+\eta\frac{D^*}{Y^*}) + \frac{D^*}{Y^*}\frac{\beta\Xi q_0}{1-\beta\lambda_s}} = \frac{\frac{\tau_y \chi}{1-\beta\lambda_s} + \frac{D^*}{Y^*}\eta}{1+\eta\frac{D^*}{Y^*} + \frac{D^*}{Y^*}\frac{\beta\Xi}{1-\beta\lambda_s}}$$

Proposition one also states $\eta = \kappa\chi/(1-\beta\lambda_s)$. Inserting:

$$\nu = \frac{\frac{\tau_y \chi}{1-\beta\lambda_s} + \frac{D^*}{Y^*}\frac{\kappa\chi}{1-\beta\lambda_s}}{1+\frac{\kappa\chi}{1-\beta\lambda_s}\frac{D^*}{Y^*} + \frac{D^*}{Y^*}\frac{\beta\Xi}{1-\beta\lambda_s}} = \frac{\frac{1}{1-\beta\lambda_s}(\tau_y\chi + \frac{D^*}{Y^*}\kappa\chi)}{1+\frac{\kappa\chi+\beta\Xi}{1-\beta\lambda_s}\frac{D^*}{Y^*}} = \frac{\frac{1}{1-\beta\lambda_s}(\tau_y\chi + \frac{D^*}{Y^*}\kappa\chi)}{\frac{1}{1-\beta\lambda_s}(1-\beta\lambda_s + (\kappa\chi+\beta\Xi)\frac{D^*}{Y^*})}$$

Inserting (123) lets us finally arrive at

$$\nu = \frac{\tau_y\chi + \frac{D^*}{Y^*}\kappa\chi}{\tau_d + \tau_y\chi - \beta\frac{D^*}{Y^*}\Xi + (\kappa\chi+\beta\Xi)\frac{D^*}{Y^*}} = \frac{(\tau_y + \frac{D^*}{Y^*}\kappa)\chi}{\tau_d + (\tau_y + \kappa\frac{D^*}{Y^*})\chi} \tag{126}$$

Now we differentiate (126) with respect to τ_d to understand its monotonicity regions. Start by defining $A = \tau_y + \kappa\frac{D^*}{Y^*}$ then by the quotient rule:

$$\frac{\partial\nu}{\partial\tau_d} = \frac{A\frac{\partial\chi}{\partial\tau_d}(\tau_d + A\chi) - (1 + A\frac{\partial\chi}{\partial\tau_d})A\chi}{(\tau_d + A\chi)^2} = \frac{A(\tau_d\frac{\partial\chi}{\partial\tau_d} - \chi)}{(\tau_d + A\chi)^2} \implies$$
$$\frac{\partial\nu}{\partial\tau_d} < 0 \iff \tau_d\frac{\partial\chi}{\partial\tau_d} < \chi \tag{127}$$

To finally prove that condition (127) in some calibrations does not hold (and thus defying monotonicity of ν in τ_d) we numerically check this over a grid of $(\phi, \psi) \in \mathbb{R}^2$ resulting in the figure 9 where the grey regions indicate that these combinations along with the general specification results in breaks of the monotonicity result of the baseline $r_t = 0$ considered by **angeletos2024deficits**. $\square$

G Hybrid model w. HtM and announced transfers

G.1 Consumption function (hybrid model)

Let $\mu \in [0, 1)$ denote the share of non-saving households with consumption denoted c_t^H, then the aggregate consumption is

$$c_t = \mu c_t^H + (1 - \mu) c_t^O, \tag{128}$$

In section A.1 we derived the OLG-households linearized consumption function for households able to save (61). Aggregating this over the share of savers $1 - \mu$ we obtain

$$c_t^O = (1 - \beta\omega)\left(\tilde{a}_t^O + \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})\right) - \xi \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k r_t \tag{129}$$

Because of monotone preferences we have that the non-savers (sometimes called Hand to Mouth consumers) consume their total disposable income in a given period t and

$$c_t^H = y_t - t_t. \tag{130}$$

This setup also implies that public debt d_t are held by the share of saving households such that

$$(1 - \mu)\tilde{a}_t^S = d_t. \tag{131}$$

Combining (129), (130) and (131) with (128) yields the linearized consumption function of this model

$$\begin{aligned}
c_t &= \mu(y_t - t_t) + (1 - \mu)\left((1 - \beta\omega)\left(\tilde{a}_t^O + \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})\right) - \xi \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k r_t\right) \\
&= (1 - \beta\omega) d_t + \mu(y_t - t_t) + (1 - \mu)\left((1 - \beta\omega)\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k}) - \xi \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k r_t\right) \\
&= (1 - \beta\omega) d_t + (\mu + (1 - \mu)(1 - \beta\omega))(y_t - t_t) \\
&\qquad + (1 - \mu)(1 - \beta\omega)\left(\mathbb{E}_t \sum_{k=1}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})\right) - (1 - \mu)\xi \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k r_t
\end{aligned}$$

Collecting terms we obtain

$$c_t = M_d d_t + M_y\left((y_t - t_t) + \delta \mathbb{E}_t \sum_{k=1}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})\right) - M_r \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k r_t \tag{132}$$

where $M_d \equiv 1 - \beta\omega$, $M_y = \mu + (1 - \mu)(1 - \beta\omega)$, $\delta \equiv \frac{(1-\mu)(1-\beta\omega)}{\mu+(1-\mu)(1-\beta\omega)}$ and $M_r \equiv (1 - \mu)\xi$.

G.2 Aggregate demand (hybrid model)

Recall that taking period t expectations of the debt transition equation (17) at time $t+k$ for $k \geq 1$ yields

$$\mathbb{E}_t[d_{t+k}] = \mathbb{E}_t\left[\frac{D^*}{Y^*}r_{t+k-1} + \frac{1}{\beta}\Big((1-\tau_d)d_{t+k-1} - \tau_y y_{t+k-1} + (1-\tau_d)e_{t+k-1}\Big)\right]. \quad (133)$$

Because the HtM-extension separates out the current disposable income term $y_t - t_t$, the relevant discounted tax-flow object is now the future series starting at $k = 1$. Hence

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k d_{t+k}\right] &= \frac{D^*}{Y^*}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k r_{t+k-1}\right] + \frac{1-\tau_d}{\beta}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k d_{t+k-1}\right] \\
&\quad + \frac{1-\tau_d}{\beta}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k-1}\right] - \frac{\tau_y}{\beta}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k-1}\right] \\
&= \beta\omega\frac{D^*}{Y^*}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right] + \omega(1-\tau_d)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k d_{t+k}\right] \\
&\quad + \omega(1-\tau_d)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k e_{t+k}\right] - \omega\tau_y\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k y_{t+k}\right].
\end{aligned} \quad (134)$$

Using $\mathbb{E}_t\big[\sum_{k=0}^{\infty}(\beta\omega)^k d_{t+k}\big] = d_t + \mathbb{E}_t\big[\sum_{k=1}^{\infty}(\beta\omega)^k d_{t+k}\big]$, and similarly for y_t and e_t, this can be rearranged to

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k d_{t+k}\right] &= \frac{\omega(1-\tau_d)}{1-\omega(1-\tau_d)}(d_t + e_t) - \frac{\omega\tau_y}{1-\omega(1-\tau_d)}y_t \\
&\quad + \frac{\omega(1-\tau_d)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
&\quad - \frac{\omega\tau_y}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
&\quad + \frac{\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right].
\end{aligned} \quad (135)$$

Using the tax rule $t_t = \tau_y y_t + \tau_d(d_t + e_t) - e_t$, it follows that

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k t_{t+k}\right] &= \tau_y \mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] + \tau_d \mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k d_{t+k}\right] \\
&\qquad - (1-\tau_d)\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
&= \frac{\tau_d \omega(1-\tau_d)}{1-\omega(1-\tau_d)}(d_t + e_t) - \frac{\tau_d \omega \tau_y}{1-\omega(1-\tau_d)} y_t \\
&\quad + \left(\tau_y - \frac{\tau_d \omega \tau_y}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
&\quad + \left(\frac{\tau_d \omega(1-\tau_d)}{1-\omega(1-\tau_d)} - (1-\tau_d)\right)\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
&\quad + \frac{\tau_d \beta\omega \frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right]. \qquad (136)
\end{aligned}$$

Reducing the coefficients as in section D.1 gives

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k t_{t+k}\right] &= \frac{\tau_d \omega(1-\tau_d)}{1-\omega(1-\tau_d)}(d_t + e_t) - \frac{\tau_d \omega \tau_y}{1-\omega(1-\tau_d)} y_t \\
&\quad + \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
&\quad - \frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
&\quad + \frac{\tau_d \beta\omega \frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right]. \qquad (137)
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k (y_{t+k} - t_{t+k})\right] &= \left(1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
&\quad + \frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
&\quad - \frac{\tau_d \omega(1-\tau_d)}{1-\omega(1-\tau_d)}(d_t + e_t) + \frac{\tau_d \omega \tau_y}{1-\omega(1-\tau_d)} y_t \\
&\quad - \frac{\tau_d \beta\omega \frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right]. \qquad (138)
\end{aligned}$$

Substituting $y_t - t_t = y_t - (\tau_y y_t + \tau_d(d_t + e_t) - e_t) = (1-\tau_y)y_t - \tau_d d_t + (1-\tau_d)e_t$, and the expression for discounted disposable income flows (138) into the linearized consumption function (132) and imposing market clearing $y_t = c_t$, yields

$$\begin{aligned}
y_t = {} & M_d d_t + M_y\Big((1-\tau_y)y_t - \tau_d d_t + (1-\tau_d)e_t\Big) \\
& + M_y\delta\left(1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
& + M_y\delta\frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
& - M_y\delta\frac{\tau_d\omega(1-\tau_d)}{1-\omega(1-\tau_d)}(d_t + e_t) \\
& + M_y\delta\frac{\tau_d\omega\tau_y}{1-\omega(1-\tau_d)}y_t \\
& - \left(M_r + M_y\delta\frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right].
\end{aligned} \tag{139}$$

Collecting terms in y_t on the left-hand side gives

$$\begin{aligned}
& \left[1 - M_y(1-\tau_y) - M_y\delta\frac{\tau_d\omega\tau_y}{1-\omega(1-\tau_d)}\right]y_t \\
& \quad = \left[M_d - M_y\tau_d - M_y\delta\frac{\tau_d\omega(1-\tau_d)}{1-\omega(1-\tau_d)}\right]d_t \\
& \quad + \left[M_y(1-\tau_d) - M_y\delta\frac{\tau_d\omega(1-\tau_d)}{1-\omega(1-\tau_d)}\right]e_t \\
& \quad + M_y\delta\frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] \\
& \quad + M_y\delta\left(1 - \frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] \\
& \quad - \left(M_r + M_y\delta\frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}\right)\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right].
\end{aligned} \tag{140}$$

Define

$$\Omega_\mu \equiv 1 - M_y\left((1-\tau_y) + \delta\frac{\tau_d\omega\tau_y}{1-\omega(1-\tau_d)}\right). \tag{141}$$

Then the AD-curve for the hybrid model with the full expected path of deficit shocks is

$$y_t = \mathrm{X}^{\mu}_{d} d_t + \mathrm{X}^{\mu}_{e,0} e_t + \mathrm{X}^{\mu}_{e,+}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k e_{t+k}\right] + \mathrm{X}^{\mu}_{y}\mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] - \mathrm{X}^{\mu}_{r}\mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right], \tag{142}$$

where

$$\mathrm{X}_d^\mu \equiv \frac{M_d - M_y\tau_d - M_y\delta\frac{\tau_d\omega(1-\tau_d)}{1-\omega(1-\tau_d)}}{\Omega_\mu}, \quad \mathrm{X}_{e,0}^\mu \equiv \frac{M_y(1-\tau_d) - M_y\delta\frac{\tau_d\omega(1-\tau_d)}{1-\omega(1-\tau_d)}}{\Omega_\mu},$$

$$\mathrm{X}_{e,+}^\mu \equiv \frac{M_y\delta\frac{(1-\tau_d)(1-\omega)}{1-\omega(1-\tau_d)}}{\Omega_\mu}, \quad \mathrm{X}_y^\mu \equiv \frac{M_y\delta\left(1-\frac{\tau_y(1-\omega)}{1-\omega(1-\tau_d)}\right)}{\Omega_\mu}, \quad \mathrm{X}_r^\mu \equiv \frac{M_r + M_y\delta\frac{\tau_d\beta\omega\frac{D^*}{Y^*}}{1-\omega(1-\tau_d)}}{\Omega_\mu}.$$

G.3 Homogeneous system of equations for determinacy calculations

To calculate which calibration admits a unique bounded equilibrium path I derive the homogeneous system of equations governing the evolution of the endogenous state variables.

Note that the BK-condition needs to hold for A ind

$$\mathbb{E}_t[X_{t+1}] = AX_t + B\epsilon_t \tag{143}$$

where ϵ_t is a vector of exogenous variables (here only containing deficit shocks, e_t. Therefore I set the exogenous path to zero ($e_t = 0,\ \forall t$) and derive the homogeneous system. Then the HtM DIS curve (142) reduces to

$$y_t = \mathrm{X}_d^\mu d_t + \mathrm{X}_y^\mu \mathbb{E}_t\left[\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k}\right] - \mathrm{X}_r^\mu \mathbb{E}_t\left[\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}\right], \tag{144}$$

and define $S_t^y \equiv \mathbb{E}_t\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+k},\ S_t^r \equiv \mathbb{E}_t\sum_{k=0}^{\infty}(\beta\omega)^k r_{t+k}$. Note that

$$\begin{aligned}S_t^y &= \beta\omega\mathbb{E}_t[y_{t+1}] + \mathbb{E}_t\sum_{k=2}^{\infty}(\beta\omega)^k y_{t+k} = \beta\omega\mathbb{E}_t[y_{t+1}] + \beta\omega\mathbb{E}_t\sum_{k=1}^{\infty}(\beta\omega)^k y_{t+1+k}\\ &= \beta\omega\mathbb{E}_t[y_{t+1}] + \beta\omega\mathbb{E}_t[S_{t+1}^y],\end{aligned}$$

and $S_t^r = r_t + \beta\omega\mathbb{E}_t[S_{t+1}^r]$ as shown multiple times before. Therefore we can rewrite the DIS curve:

$$\begin{aligned}\mathbb{E}_t y_{t+1} &= \mathrm{X}_d^\mu\mathbb{E}_t d_{t+1} + \mathrm{X}_y^\mu\mathbb{E}_t S_{t+1}^y - \mathrm{X}_r^\mu\mathbb{E}_t S_{t+1}^r \iff\\ \beta\omega\mathbb{E}_t y_{t+1} &= \beta\omega\mathrm{X}_d^\mu\mathbb{E}_t d_{t+1} + \beta\omega\mathrm{X}_y^\mu\mathbb{E}_t S_{t+1}^y - \beta\omega\mathrm{X}_r^\mu\mathbb{E}_t S_{t+1}^r\\ &= \beta\omega\mathrm{X}_d^\mu\mathbb{E}_t d_{t+1} + \mathrm{X}_y^\mu(S_t^y - \beta\omega\mathbb{E}_t y_{t+1}) - \mathrm{X}_r^\mu(S_t^r - r_t) \iff\\ (1+\mathrm{X}_y^\mu)\beta\omega\mathbb{E}_t y_{t+1} &= \beta\omega\mathrm{X}_d^\mu\mathbb{E}_t d_{t+1} + \mathrm{X}_y^\mu S_t^y - \mathrm{X}_r^\mu S_t^r + \mathrm{X}_r^\mu r_t\end{aligned}$$

from here insert $\mathrm{X}_y \mu S_t^y - \mathrm{X}_r^\mu S_t^r = y_t - \mathrm{X}_d^\mu d_t$:

$$
\begin{aligned}
(1+\mathrm{X}_y^\mu)\beta\omega \mathbb{E}_t y_{t+1} &= \beta\omega \mathrm{X}_d^\mu \mathbb{E}_t d_{t+1} + y_t - \mathrm{X}_d^\mu d_t + \mathrm{X}_r^\mu r_t \\
&= \mathrm{X}_d^\mu \left[\omega(1-\tau_d) - 1\right] d_t + \left[1 - \omega \mathrm{X}_d^\mu \tau_y + \beta\omega \mathrm{X}_d^\mu \frac{D^*}{Y^*}\alpha_y + \mathrm{X}_r^\mu \alpha_y\right] y_t + \\
&\qquad \left[\beta\omega \mathrm{X}_d^\mu \frac{D^*}{Y^*}\alpha_\pi + \mathrm{X}_r^\mu \alpha_\pi\right] \pi_t
\end{aligned}
$$

where we have used the debt law (146).

Combining the monetary policy and the Phillips curve yields (as in section E)

$$
r_t = \left(\phi + \frac{\kappa}{\beta}\right) y_t + \left(\psi - \frac{1}{\beta}\right)\pi_t = \alpha_y y_t + \alpha_\pi \pi_t, \tag{145}
$$

where $\alpha_y \equiv \phi + \kappa\beta^{-1}$ and $\alpha_\pi \equiv \psi - \beta^{-1}$.

And the debt law of motion (17) reduces to

$$
\begin{aligned}
\mathbb{E}_t[d_{t+1}] &= \frac{1-\tau_d}{\beta} d_t - \frac{\tau_y}{\beta} y_t + \frac{D^*}{Y^*} r_t \\
&= \frac{1-\tau_d}{\beta} d_t + \left(\frac{D^*}{Y^*}\alpha_y - \frac{\tau_y}{\beta}\right) y_t + \frac{D^*}{Y^*}\alpha_\pi \pi_t
\end{aligned} \tag{146}
$$

Combined we have

$$
\mathbb{E}_t \begin{bmatrix} d_{t+1} \\ y_{t+1} \\ \pi_{t+1} \end{bmatrix} = \begin{bmatrix} m_{d,d}^\mu & m_{d,y}^\mu & m_{d,\pi}^\mu \\ m_{y,d}^\mu & m_{y,y}^\mu & m_{y,\pi}^\mu \\ 0 & -\frac{\kappa}{\beta} & \frac{1}{\beta} \end{bmatrix} \begin{bmatrix} d_t \\ y_t \\ \pi_t \end{bmatrix}, \tag{147}
$$

where

$$
m_{d,d}^\mu \equiv \frac{1-\tau_d}{\beta},\ m_{d,y}^\mu \equiv \frac{D^*}{Y^*}\left(\phi + \frac{\kappa}{\beta}\right) - \frac{\tau_y}{\beta},\ m_{d,\pi}^\mu \equiv \frac{D^*}{Y^*}\left(\psi - \frac{1}{\beta}\right).
$$

$$
\begin{aligned}
m_{y,d}^\mu &\equiv \frac{-\mathrm{X}_d^\mu\left[1-\omega(1-\tau_d)\right]}{\beta\omega\left(1+\mathrm{X}_y^\mu\right)},\ m_{y,y}^\mu \equiv \frac{1-\omega \mathrm{X}_d^\mu \tau_y + \left[\beta\omega \mathrm{X}_d^\mu \frac{D^*}{Y^*} + \mathrm{X}_r^\mu\right]\left(\phi + \frac{\kappa}{\beta}\right)}{\beta\omega\left(1+\mathrm{X}_y^\mu\right)}, \\
m_{y,\pi}^\mu &\equiv \frac{\left[\beta\omega \mathrm{X}_d^\mu \frac{D^*}{Y^*} + \mathrm{X}_r^\mu\right]\left(\psi - \frac{1}{\beta}\right)}{\beta\omega\left(1+\mathrm{X}_y^\mu\right)}.
\end{aligned}
$$

This matrix of the homogenous HtM-OLG system is used when numerically solving the model - to check if a given parametrisation have a unique bounded equilibrium.

H Alternative calibrations

H.1 No HtM share

This section of the appendix present the IRF plots of section 5 where $\mu = 0$.

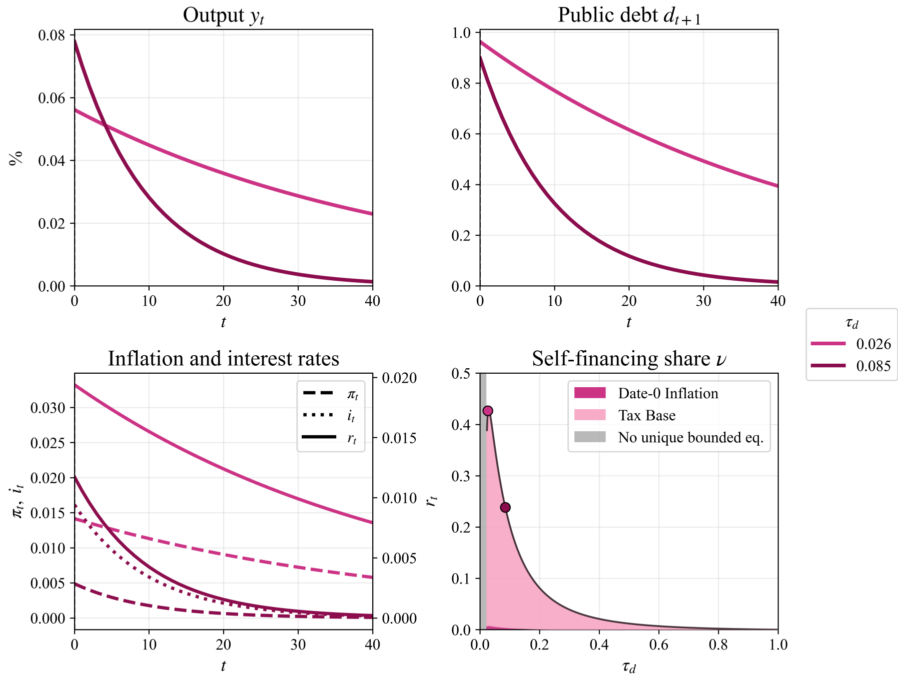

Figure 10: IRFs for calibration with $r_t = 0.0862 y_t + 1.0275 \pi_t$ and $\mu = 0$

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.

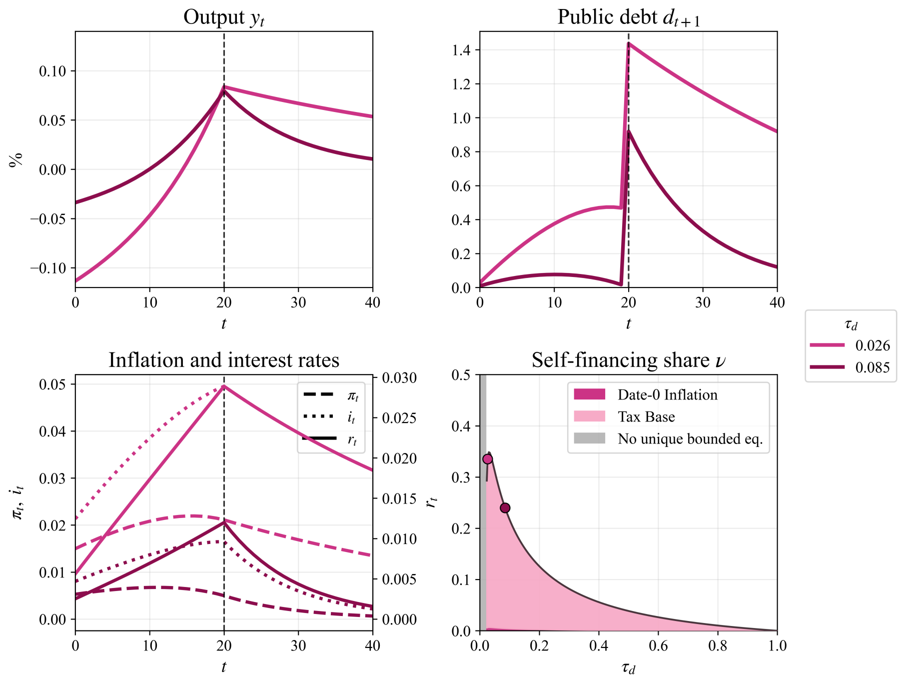

Figure 11: IRFs for announced transfer with $r_t = 0.0862y_t + 1.0275\pi_t$ and $\mu = 0$

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.

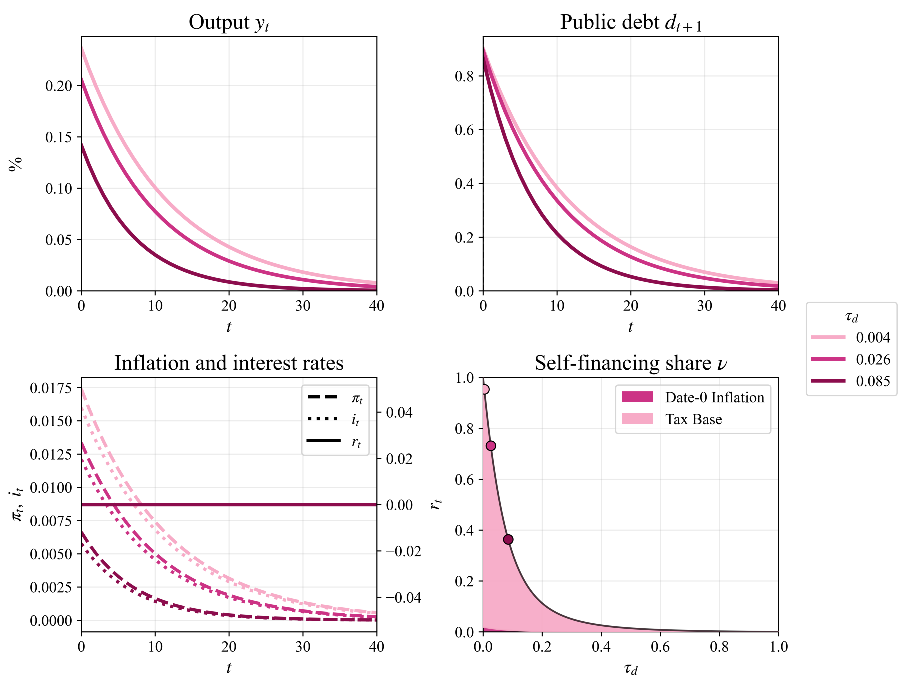

Figure 12: IRFs for calibration with $r_t = 0$ and $\mu = 0$

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.

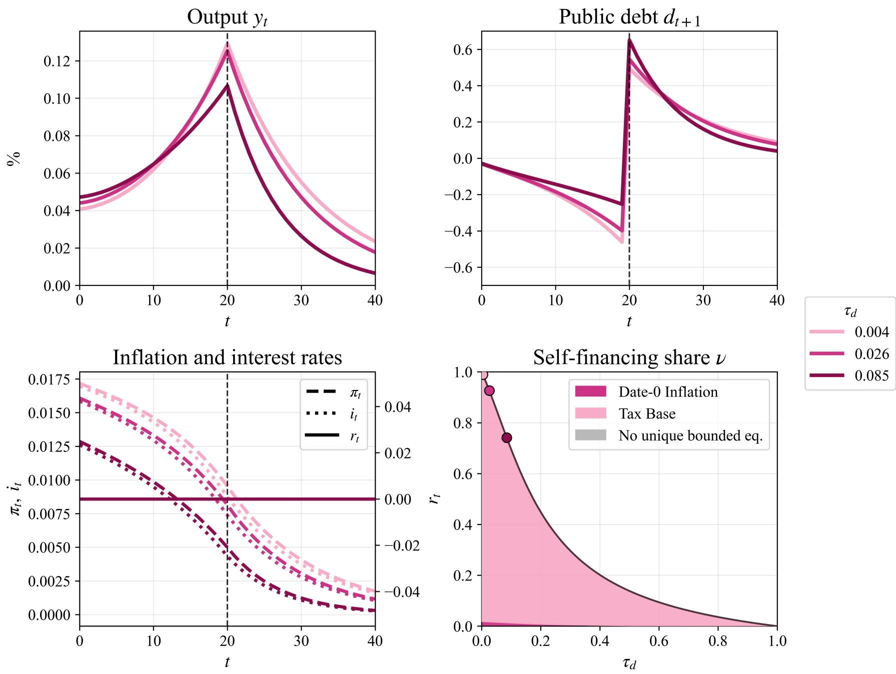

Figure 13: IRFs for announced transfer with $r_t = 0$ and $\mu = 0$

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.

H.2 Alternative real rate

This section of the appendix present the IRF plots where the real rate follows

$$r_t = \left(0.5 + \frac{\kappa}{\beta}\right) y_t + \left(1.5 - \frac{1}{\beta}\right) \pi_t,$$

where the model is otherwise calibrated as in table 1.

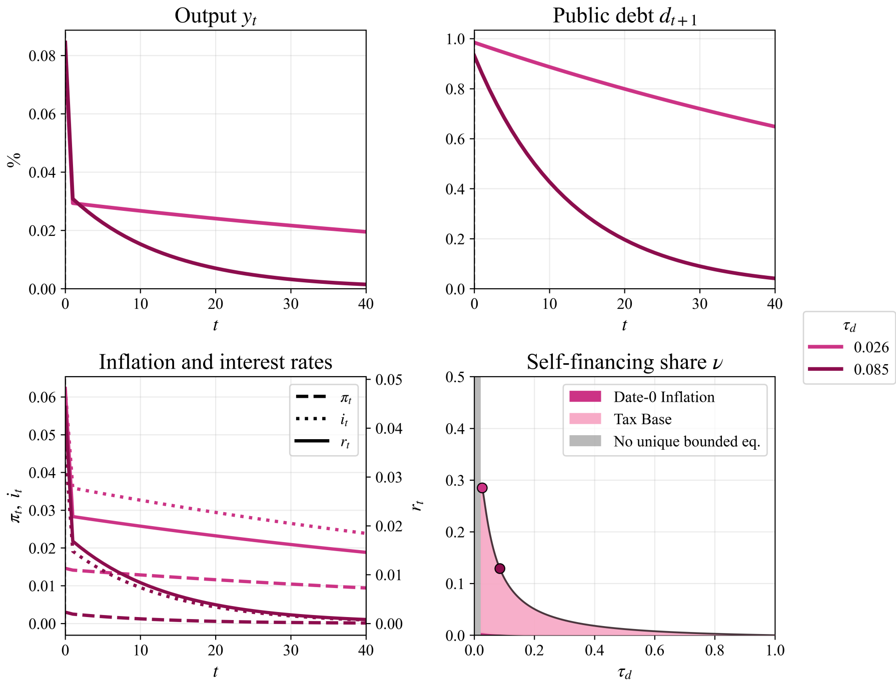

Figure 14: Impulse response functions for a fiscal deficit shock

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.

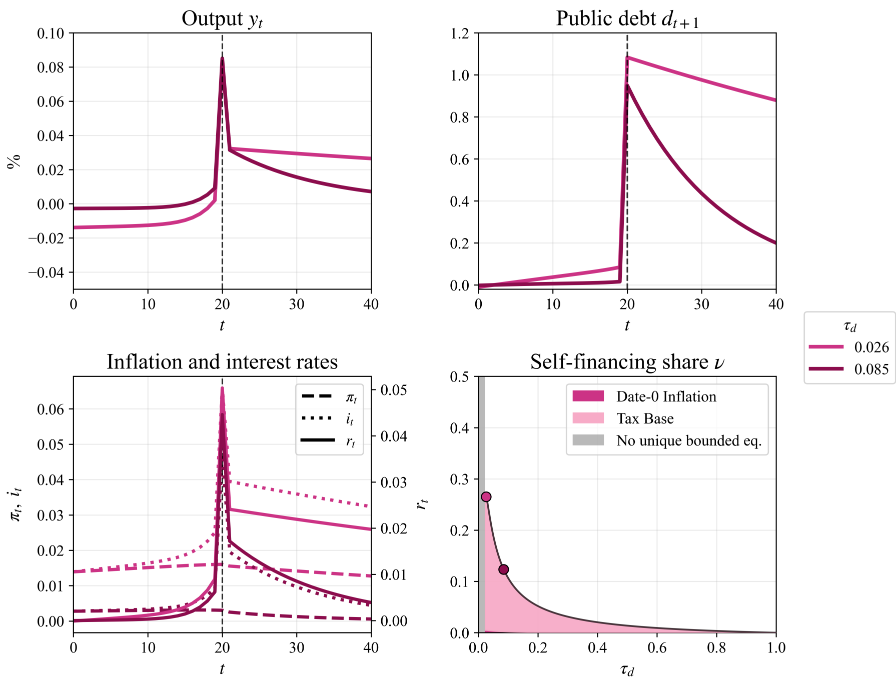

Figure 15: Impulse response functions for a fiscal deficit shock

Note: The first three panels show IRFs to a deficit shock of 1 pct. of SS output. The final panel plots ν over $\tau_d \in (0, 1)$.